

[R22] IMPORTANT QUESTIONS WITH ANSWERS

ELECTRONIC DEVICES AND CIRCUITS

UNIT 1

1. How a diode is used as switch and define all switching times.
2. Derive the expression for transition capacitance of a diode
3. Explain the formation of depletion region in a PN junction.
4. Derive the expression for diffusion capacitance of a diode.
5. Draw and explain V-I Characteristics Diode

UNIT 2

1. Compare FWR and Bridge rectifier
2. What is meant by ripple factor and derive the expression for HWR
3. A diode whose internal resistance is 20Ω is to supply power to a 100Ω load from $110V$ (R.M.S) source of supply. Calculate: i) Peak Load Current ii) DC Load Current iii) AC Load Current iv) % Regulation from No load to given load.
4. Explain the operation of Full Wave Rectifier with necessary graphs.
5. Derive the expression for Ripple factor for Full Wave Rectifier with L-section filter.
6. With a neat circuit diagram, Explain the operation of Full Wave Rectifier with Capacitor filter.
7. A Half Wave Rectifier circuit supplies 100 mA DC current to a 250Ω load. Find the DC output voltage, PIV rating of the diode and rms voltage for the transformer supplying the rectifier
8. State and prove Clamping Circuit Theorem.
9. Design a clipping circuit that clips at two independent levels.
10. In half-wave rectifier an ac voltage of peak value $24V$ is connected in series with silicon diode and load resistance of 480Ω . If the forward resistance of the diode is 20Ω , find average load current and rms value of load current.
11. Explain the operation of negative clamping circuit.

UNIT 3

1. a) Explain about input and output characteristics of a transistor when it is Connected in common base configuration.
b) Determine I_C , I_E and α for a transistor circuit having $I_B=15\mu A$ and $\beta=150$.
2. a) Draw and explain the CE characteristics of a transistor.
b) The reverse leakage current of the transistor when in CB configuration is $0.3\mu A$ while it is $16\mu A$ when the same transistor is connected in CE configuration. Determine α , β and γ .
3. Explain the working of the transistor as a switch.
4. a) Explain input and output characteristics of a transistor in CC configuration with neat sketches. Common Collector Configuration:
b) A transistor has a typical β of 100. If the collector current is 40 mA, what is the value of emitter current?
5. a) Determine I_C , I_E and α for a transistor circuit having $I_B=15\mu A$ and $\beta=150$.
b) With necessary circuit and waveform, explain the switching characteristics of a transistor in detail.
6. Explain principal of operation of BJT

UNIT4

1. Explain the construction of JFET and write down principle of operation of JFET?
2. Why we call FET as a Voltage Controlled Device
3. a) A FET follows the relation $I_D = I_{DSS} (1 - V_{GS}/V_P)^2$. What are the values of I_D and g_m for $V_{GS} = -1.5 V$ if I_{DSS} and V_P are given as 8.4 mA and -3 V respectively?
b) State any three differences between JFET and BJT.
4. a) Explain the construction, principle of operation and Volt-Ampere characteristics of JFET.
b) A certain JFET operates in the linear region with a constant drain voltage of 1V. When the gate voltage is 2V, a drain current of 10mA flows, but when gate voltage is changed to 1V, the drain current becomes 22.8mA.
5. a) Differentiate between a BJT and FET.
b) Explain the operation of FET with its characteristics and explain the different regions in transfer characteristics.
6. Explain the construction of MOSFET and write down principle of operation of JFET?

7. Explain MOSFET as a Capacitor.

UNIT 5

1. a) Describe the construction and working of UJT with its equivalent.
b) With a neat diagram explain the working of Varactor Diode
2. a) Explain the working principle of Tunnel diode and explain how it works as an Oscillator?
b) Explain how the zener diode is used for regulation purpose.
3. a) Draw the circuit diagram of SCR and explain its operation along with its characteristics.
b) With a neat diagram explain the working and characteristics of Zener Diode.
4. a) Draw the circuit diagram of photo diode and explain its operation along with its characteristics.
b) Draw the circuit diagram of Schottky diode and explain its operation along with its characteristics.
5. Explain about LED.

UNIT 1

1. How a diode is used as switch and define all switching times.

PN Diode Switching Times:

A P-N junction diode may be used as an electrical switch. The electrical circuit can be made 'on' and 'off' by forward biasing or reverse biasing the diode. In both the cases, the diode response is accompanied by transient, and an interval of time elapses before the diode recovers to its steady state. The duration of time elapsed before the recovery of the steady state of the diode is called the recovery time, t_r . On PN Diode Switching Times from reverse-bias condition to forward-bias condition, the diode takes certain time interval, called the forward recovery time, t_{fr} . The forward recovery time t_{fr} is defined as the time interval between the instant of 10% diode voltage and the instant this voltage attains 90% of its final value. It is found that t_{fr} does not usually constitute a serious practical problem and so only the more important situation of reverse recovery will be discussed here.

Diode Reverse Recovery Time

As already discussed, when a P-N junction is forward biased a large number of electrons cross over from N-region to P-region and similarly a large number of holes cross over from P-region to N-region. The electrons in the P-type region and the holes in the N-type region establish a large number of minority carriers in each region. The variation of steady-state minority carrier density with distance x from the junction in a forward biased P-N junction is shown in Fig. 7.73 (a). The number of minority carriers is very large. These minority carriers have, in each case, been supplied from the other side of the junction, where, being majority carriers, they are in plentiful supply. The concentration of holes in N-region, P_n is increased above its thermal-equilibrium value P_{no} . Similarly the concentration of electrons in P-region, n_p is increased above its thermal-equilibrium value n_{po} . The difference of excess of holes i.e., $(P_n - P_{no})$ and excess of electrons i.e., $(n_p - n_{po})$ is also shown in the figure.

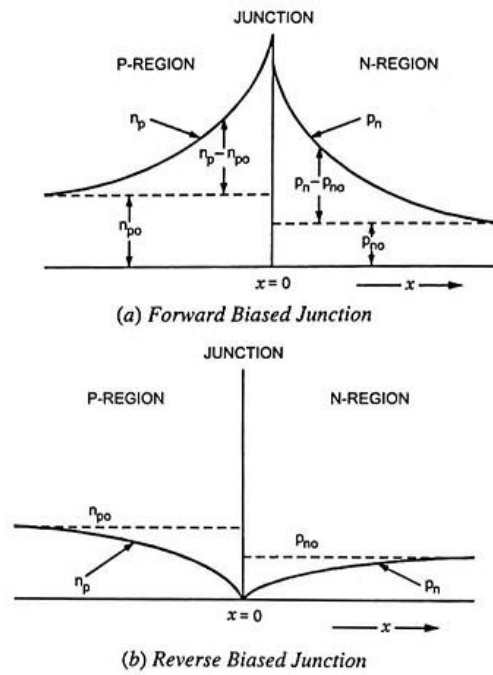


Fig. 7.73 Minority-Carrier Density Distribution as a Function of Distance x from The Junction

When the junction is reverse biased, the steady-state density of minority carriers is as shown in Fig. 7.73 (b). Far from the junction the minority carriers are equal to their thermal-equilibrium values p_{no} and n_{po} as is also the situation in Fig. 7.73 (a). As the minority carriers approach the junction, they are rapidly swept across, and the density of minority carriers diminishes to zero at this junction. The reverse saturation current I_0 is very small because the density of the thermally generated minority carriers is very small.

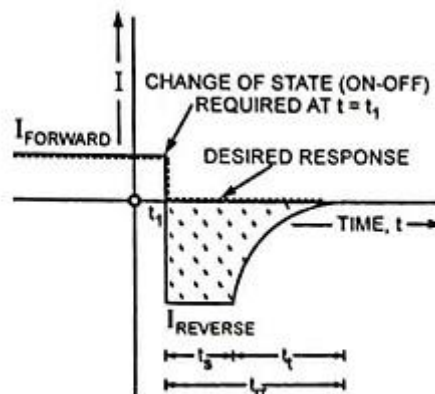


Fig. 7.74

If the external voltage is suddenly reversed in a diode circuit which has been carrying current in the forward direction, the diode current will not immediately fall to its steady-state reverse voltage value, as illustrated in Fig. 7.74. However, because of the large number of minority carriers in both of

the regions, the diode current will simply reverse, as illustrated in Fig. 7.74 and stay at this measurable level required for the minority carriers to reduce to the distribution shown in Fig. 7.73 (b) from the minority-carrier distribution shown in Fig. 7.73 (a). The diode will continue to conduct for a time, known as diode reverse recovery time, t_{rr} , until excess minority-carrier density $P_n - P_{n0}$ or $(n_p - n_{p0})$ has dropped normally to zero. In this situation, the current is determined by the external resistance in the diode circuit.

Storage and Transition Times:

The sequence of events occurring on reverse biasing of a conducting diode is shown in Fig. 7.75.

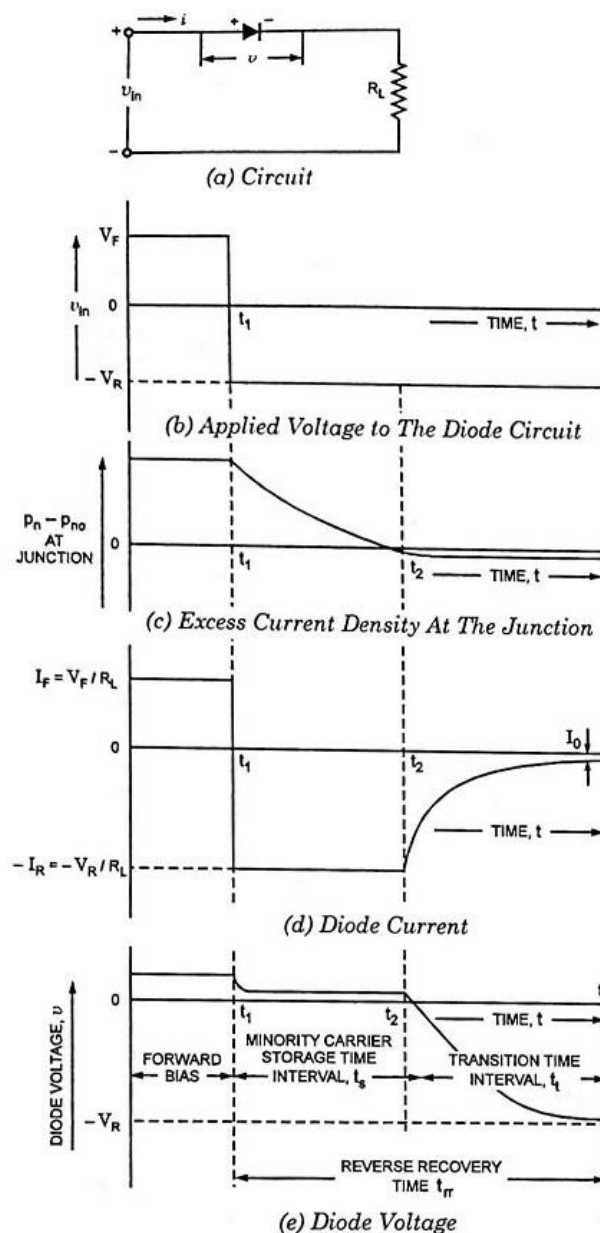


Fig. 7.75 Waveform in a Diode On Reversing of Bias

Let the input voltage v_{in} applied to a diode-resistance circuit shown in Fig. 7.75 (a) be reversed abruptly at time instant t_1 . Up to a time t_1 , the diode is conducting in forward direction and input voltage $v_{in} = V_F$. Assuming load resistance R_L to be large enough to cause voltage drop across it large in comparison with the voltage drop across the diode, the current flowing through the circuit, $i \approx V_F/R_L \approx I_F$. At instant of time t_1 , the input voltage reverses abruptly to a value $-V_R$. Now the current does not drop to zero, but instead reverses and remains at a value $i \approx -V_R/R_L \approx -I_R$ for the time duration t_s (storage time interval) required for the minority carriers to return to their majority-carrier state in the opposite region. At instant of time t_2 , as seen in Fig. 7.75 (c), the injected minority-carrier density at $x = 0$ has reached the equilibrium state.

Now if we consider the diode voltage v , we find that at time instant t_1 the diode voltage falls slightly by $(I_F + I_R) R_d$, where R_d is the diode ohmic resistance, but does not reverse. At time instant t_2 the process of sweeping of the excess minority carriers in the vicinity of junction has completed. So the diode voltage begins to reverse and the magnitude of diode current begins to decrease. During the time interval t_1 to t_2 , the stored minority charge has remained stored and, therefore, this time interval is called the storage time t_s . This is illustrated in Fig. 7.75 (e).

After all excess minority carriers are swept out or recombined in the neighborhood of junction, the diode voltage begins to reverse and the magnitude of diode current begins to fall exponentially to the normally steady-state reverse saturation current value. The time duration which elapses between t_2 and the time instant t_3 when the diode has nominally recovered, is called the transition time t_t . This recovery time interval will be completed when the minority carriers, which are at some distance from the junction, have diffused to the junction and crossed it and when, in addition, the junction transition capacitance across the reverse-biased junction has charged through load resistance R_L to the voltage $-V_R$.

The sum of the storage time t_s and transition time t_t is called the diode reverse recovery time t_{rr} i.e., $t_s + t_t = t_{rr}$.

Manufacturers normally specify the reverse recovery time of a diode t_{rr} in a typical operating condition in terms of the current waveform of Fig. 7.75 (d). The time t_{rr} is the interval from the current reversal at $t = t_1$ until the diode has recovered to a specified extent in terms either of the diode current or of the diode resistance.

Naturally t_{rr} is an important consideration in high speed PN Diode Switching Times applications. Most commercially available PN Diode Switching Times have reverse recovery time t_{rr} in the range of a few nanoseconds to 1 μ s. Units are available, however, with a t_{rr} of only a few hundred picoseconds.

2) Derive the expression for transition capacitance of a diode

Transition (or Space-Charge) Capacitance:

When a P-N junction is reverse biased, the depletion region acts like an insulator or dielectric material while the P- and N-type regions on either side have a low resistance and act as the plates. Thus P-N junction may be considered a parallel plate capacitor. The junction capacitance is termed as space-charge capacitance or transition capacitance or depletion region capacitance and is denoted by C_T .

As mentioned earlier, a reverse bias causes majority carriers to move away from the junction, thereby uncovering more immobile charges. So the thickness W of the depletion layer increases with the increase in reverse bias voltage. This increase in uncovered charge with applied voltage may be considered a capacitive effect. The incremental capacitance C_T may be defined as

$$C_T = \left| \frac{dQ}{dV} \right| \quad \dots(7.41)$$

where dQ is the increase in charge due to increase in voltage, dV .

$$i = \frac{dQ}{dt} = C_T \frac{dV}{dt} \quad \dots(7.42)$$

Since the depletion region increases with the increase in reverse-bias potential, the resulting transition capacitance decreases as shown in Fig. 7.67.

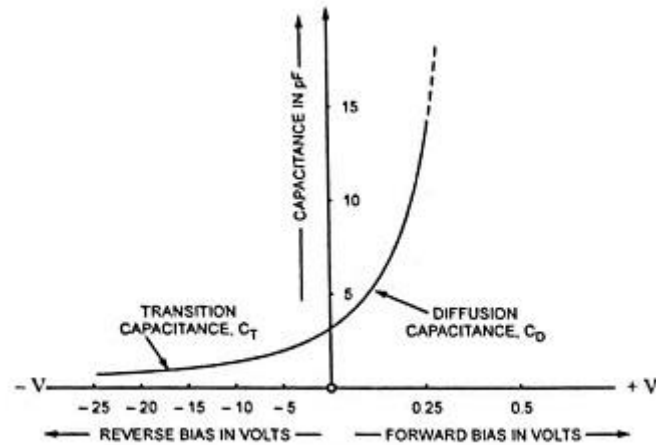


Fig. 7.67

The fact that the capacitance depends on the applied reverse bias, has application in a number of electronic systems.

1. Step Graded Junction. A junction is said to be step graded if there is an abrupt change from acceptor ion concentration on the P-side to donor ion concentration on the N-side. Such a junction gets formed in alloyed junction or fused junction diode. Usually the acceptor density N_A and the donor density N_D are kept unequal.

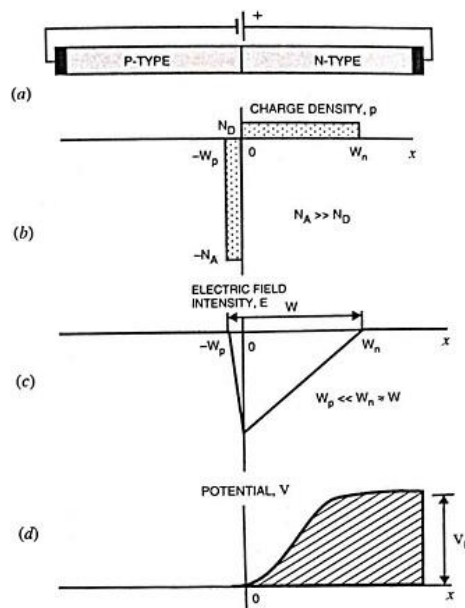


Fig. 7.68 Charge Density, Field Intensity and Potential Variation With Distance x

Figure 7.68 shows the charge density as a function of distance from an alloy junction in which acceptor impurity density is assumed to be much larger than the donor concentration. Since the net charge is zero,

$$eN_A W_p = eN_D W_n \quad \dots(7.43)$$

When N_A is much larger than N_D , the $W_p \ll W_n$ and $W_n \approx W$.

The relationship between potential and charge density is given by the Poisson's equation,

$$\frac{d^2V}{dx^2} = \frac{-eN_D}{\epsilon} \quad \dots(7.44)$$

where ϵ is the permittivity of the semiconductor and is given as $\epsilon = \epsilon_r \epsilon_0$ where ϵ_r is the relative permittivity of the dielectric and ϵ_0 is the permittivity of free space.

The electric lines of flux start on the positive donor ions and terminate on the negative acceptor ions. So there are no flux lines to the right of the boundary $x = W_n$ in Fig. 7.68

and field intensity

$$\epsilon = -\frac{dV}{dx} = 0$$

at

$$x = W_n = W$$

Integrating Eq. (7.44) subject to these boundary conditions, we have

$$\frac{dV}{dx} = \frac{-eN_D}{\epsilon}(x - W) = -\epsilon \quad \dots(7.45)$$

Neglecting the small voltage drop across W_p , we may arbitrarily choose $V = 0$ at $x = 0$

Integrating Eq. (7.45) subject to these conditions

$$V = \frac{-eN_D}{2\epsilon}(x^2 - 2Wx) \quad \dots(7.46)$$

The linear variation in field intensity and the quadratic dependence of potential upon distance are plotted in Fig. 7.68 (c) and (d).

At $x = W$, $V = V_B$, the barrier height. Thus

$$V_B = \frac{eN_D W^2}{2\epsilon} \quad \dots(7.47)$$

Here $V_B = V_0 - V$ where V is the negative number for an applied reverse bias and V_0 is the contact potential.

Hence, the width of depletion layer increases with applied reverse bias and varies as $\sqrt{V_B}$.

The total charge density of N-type material with area of junction A is given by

$$Q = eN_D WA$$

The transition capacitance C_T , given by Eq. (7.41), is

$$C_T = \left| \frac{dQ}{dV} \right| = eN_D A \left| \frac{dW}{dV_B} \right|$$

From Eq. (7.47),

$$\left| \frac{dW}{dV_B} \right| = \frac{\epsilon}{eN_D W},$$

and therefore,

$$C_T = \frac{\epsilon A}{W} \quad \dots(7.48)$$

where ϵ is the absolute permittivity of the semiconductor medium, A is the cross-sectional area of the junction and W is the width of the depletion layer and is given as

$$W^2 = \left[\frac{2\epsilon V_B}{e} \right] \left[\frac{1}{N_A} + \frac{1}{N_D} \right] \quad \dots(7.49)$$

$$\text{or } W = \sqrt{\frac{2\epsilon V_B}{eN_D}} \quad \text{when } N_A \gg N_D \quad \dots(7.50)$$

$$\text{So } C_T = \epsilon A \sqrt{\frac{eN_D}{2\epsilon V_B}} = A \sqrt{\frac{N_D}{V_B}} \cdot \sqrt{\frac{e\epsilon}{2}} \quad \dots(7.51)$$

Thus in a step graded junction, C_T is inversely proportional to the square root of the junction voltage V_B where V_B is given as

$$V_B = V_0 - V_R \quad \dots(7.52)$$

where V_R is a negative number indicating the applied reverse bias and V_0 is the contact potential.

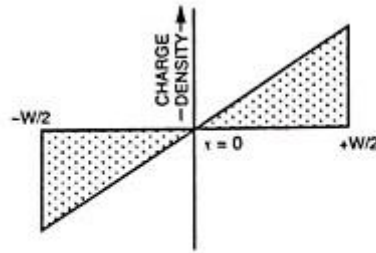


Fig. 7.69 Charge-Density Variation With Distance in a Grown P-N Junction

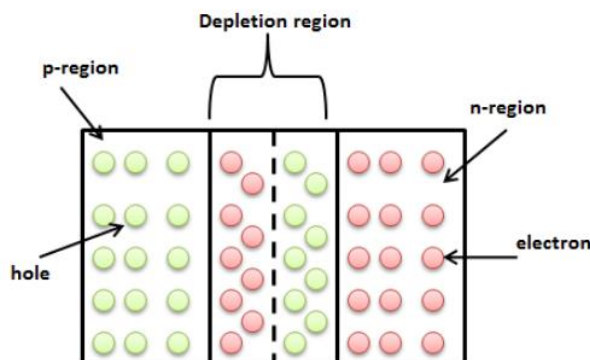
Linearly Graded Junction (or Grown Junction). A junction is said to be linearly graded if the charge density varies gradually (almost nearly) with the distance in the transition region. Such a junction gets formed in a growth junction diode.

In such a junction diode also, C_T is given by Eq. (7.48). Thus the expression for C_T for a linearly graded junction is the same as for a step graded junction. But here W is the total width of the space-charge layer. However, it now comes out that W varies as $V^{1/3}_B$ instead of $V^{1/2}_B$.

3) Explain the formation of depletion region in a PN junction.

A PN junction is a fundamental building block of semiconductor devices like diodes, transistors, and integrated circuits. It is formed by bringing together two types of semiconductors: P-type (positive charge carriers) and N-type (negative charge carriers). When these two regions are joined, they create what is known as a PN junction.

The formation of the depletion region in a PN junction is a result of the diffusion and recombination processes that occur when the two different types of semiconductors are brought into contact. Let's break down the steps of how the depletion region is formed:



1. **Doping:** The P-type semiconductor is doped with acceptor impurities (such as boron), which create holes (positively charged carriers). The N-type semiconductor is doped with donor impurities (such as phosphorus), which introduce extra electrons (negatively charged carriers).
2. **Contact:** The P-type and N-type semiconductors are physically brought into contact, forming the PN junction.
3. **Diffusion:** Due to the concentration gradient, the majority carriers (holes in the P-type and electrons in the N-type) tend to diffuse from regions of higher concentration to regions of lower concentration. As a result, holes from the P-side start diffusing towards the N-side, and electrons from the N-side start diffusing towards the P-side.
4. **Recombination:** As the holes and electrons diffuse, they come into contact with each other in the region near the junction. When a hole and an electron combine, they annihilate each other through a process called recombination. This recombination leaves behind positively charged ions on the N-side and negatively charged ions on the P-side.
5. **Electric Field:** As recombination occurs, an electric field starts building up across the depletion region due to the immobile charged ions that are left behind. This electric field opposes the further movement of holes and electrons.
6. **Depletion Region:** The region near the PN junction where most of the mobile charge carriers have diffused away or recombined is known as the depletion region. In this region, there are immobile positively charged ions on the N-side and immobile negatively charged ions on the P-side. The depletion region effectively acts as a barrier to the flow of charge carriers.

The width of the depletion region depends on the level of doping in the P and N regions and the applied voltage across the PN junction. When a forward bias voltage is applied (positive to the P-side and negative to the N-side), it reduces the width of the depletion region, allowing current to flow through the junction. In contrast, a reverse bias voltage widens the depletion region, restricting the flow of current and causing the PN junction to behave like a one-way valve, allowing current only in one direction, which is the principle behind diodes.

4) Derive the expression for diffusion capacitance of a diode.

Diffusion (or Storage) Capacitance:

When a P-N junction is forward biased, a capacitance, which is much larger than the transition capacitance, comes into play. This type of capacitance is called the diffusion capacitance, C_D . We know that for a forward-biased P-N junction, the potential barrier is reduced. Now the electrons from N-side enter the P-side and holes from P-side enter into the N-side. These charge carriers diffuse away from the junction and progressively recombine. The density of charge carriers is high near the junction and decays exponentially with distance. Thus a charge is stored on both sides of the junction when forward biased. It is observed that the amount of stored charge varies with the applied potential as for a true capacitor. It is convenient to introduce an incremental capacitance, called diffusion or storage capacitance C_D given by the equation $C_D = dQ/dV$ where dQ represents the change in number of minority carriers stored outside the depletion region when a change in voltage across diode, dV , is applied.

If τ is the mean lifetime of charge carriers, then a flow of charge Q yields a diode current I given as

$$I = \frac{Q}{\tau} \quad \text{or} \quad Q = \tau I$$

and from Eq. (7.20),

$$I = I_0 (e^{V/\eta V_T} - 1)$$

So

$$Q = \tau I_0 (e^{V/\eta V_T} - 1) = \tau I_0 e^{V/\eta V_T}$$
$$\because e^{V/\eta V_T} \gg 1$$

So diffusion capacitance

$$C_D = \frac{dQ}{dV} = \frac{d(\tau I_0 e^{V/\eta V_T})}{dV} = \frac{\tau I_0}{\eta V_T} e^{V/\eta V_T} = \frac{\tau(I + I_0)}{\eta V_T}$$

For a forward bias greater than a few tenths of a volt,

$$\left[\frac{V}{\eta V_T} \right] \gg 1$$

and according to Eq. (7.20) $I \gg I_0$ and so the diffusion capacitance is given as

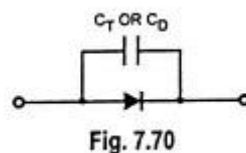
$$C_D = \frac{\tau I}{\eta V_T} \quad \dots(7.53)$$

Thus the diffusion capacitance is directly proportional to the forward current through the diode. This relationship occurs since the number of majority carriers injected into the depletion region is directly proportional to the forward current of the diode. The variation in diffusion capacitance with the variation in forward bias applied across a P-N junction is shown in Fig. 7.67.

This capacitance assumes great significance in the operation of devices which are required to switch rapidly from forward bias to reverse bias. Large value of C_D , delays both of the switching-on and the switching-off.

In fact in a forward-biased diode, both the diffusion capacitance C_D and the transition capacitance C_T are present, but $C_D \gg C_T$. Typically C_D is more than a million times greater than C_T . So in a forward-biased diode C_T may be neglected and only C_D may be considered. Typical value of C_D is $0.02\mu\text{F}$.

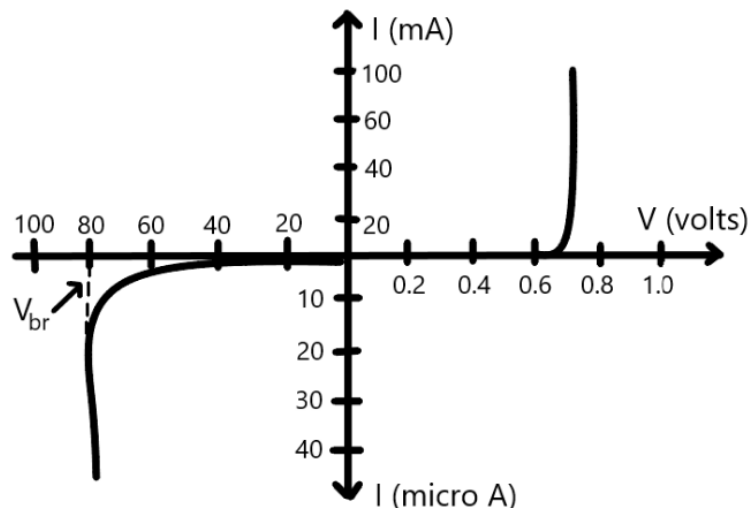
Similarly in a reverse-biased diode both C_D and C_T are present but $C_T \gg C_D$. Hence in a reverse-biased diode C_D is neglected and only C_T is considered.



5) Draw and explain V-I Characteristics Diode

Hint: The V-I characteristics of a p-n junction diode can be used to find the relation between voltage and current. The relationship between the voltage and the current will be non-linear. The minority carriers in the junction will result in the current.

The V-I characteristics of p-n junction diode represents the relation between current and voltage and the variations in current with respect to current is shown below;



Here, the X-axis represents the voltage or potential and the Y-axis represents the current in the p-n junction diode.

Now, when the p-n junction diode is reverse biased, the p-type of the semiconductor diode is connected to the negative terminal of the battery and the n-type of the semiconductor diode is connected to the positive terminal of the battery. Hence, the holes in the p-type of the semiconductor will be attracted by the negative terminal of the battery.

Also, the electrons in the n-type of the semiconductor will be attracted to the positive terminal of the battery. This results in barrier potential and hence, results in the increase of the depletion layer. As a result of this, the minority charge carrier will get drifted across the junction and result in the production of a small current.

Now, at the normal temperature, the current will be very low. Also, the applied voltage will provide very little change in the current which is almost negligible. Therefore, we can say that the current under reverse biased condition will be almost independent of the applied voltage.

Note: In the case of reverse biasing of the p-n junction diode, the built-in electric field and the applied electric field will be in the same direction. When we will add both the electric fields, the resultant electric field will be the same as that of the built-in electric field. This results in an increase in resistance in the diode and also an increase in the depletion layer of the diode.

UNIT 2

1) Compare FWR and Bridge rectifier

Here's a comparison between Full Wave Rectifier (FWR) and Bridge Rectifier in tabular form:

Aspect	Full Wave Rectifier (FWR)	Bridge Rectifier
Operation	Utilizes both halves of the input AC waveform	Utilizes both halves of the input AC waveform
Diodes Required	Requires at least two diodes	Requires at least four diodes
Voltage Polarity	Produces only positive half-wave rectification	Produces both positive and negative half-wave rectification
Efficiency	Relatively lower compared to Bridge Rectifier	Relatively higher compared to Full Wave Rectifier
Transformer Requirement	Requires a center-tapped transformer for step-down applications	Can work with a center-tapped or non-center-tapped transformer
Output Ripple	Output has higher ripple due to only half-wave rectification	Output has lower ripple due to full-wave rectification

Aspect	Full Wave Rectifier (FWR)	Bridge Rectifier
Voltage Conversion Efficiency	Lower due to higher ripple and single polarity output	Higher due to lower ripple and both polarities output
P-N Junction Stress	Diodes experience higher stress due to higher current flow	Diodes experience lower stress due to lower current flow
Cost	Relatively lower cost due to fewer diodes required	Relatively higher cost due to more diodes required
Applications	Suitable for low-power applications and low voltage levels	Suitable for a wide range of applications and voltage levels

It's worth noting that both Full Wave Rectifier (FWR) and Bridge Rectifier are widely used in various applications, and the choice between them depends on factors such as the required voltage level, current capacity, cost, and efficiency.

2) What is meant by ripple factor and derive the expression for HWR

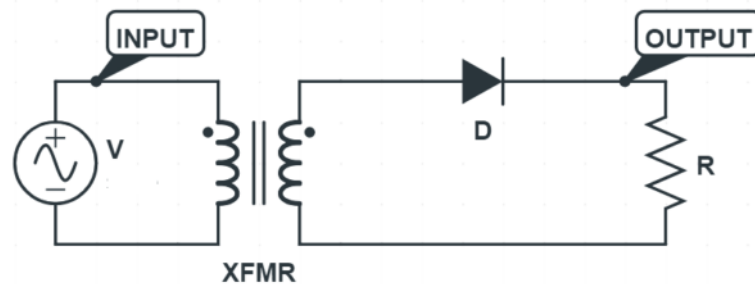
Ripple Factor of Half Wave Rectifier

The value of ripple factor of single phase half wave rectifier is equal to 1.21. Since, ripple factor is the ratio of rms value of fluctuating ac component to the average value or dc value. Its value of 1.21 means that, the ac fluctuating component in the rectified output of half wave rectifier is 121%

of the expected DC output or average value. This simply means, the rectification is poor as the ripple content is even higher than the DC output. Let us now see the calculation of this ripple factor.

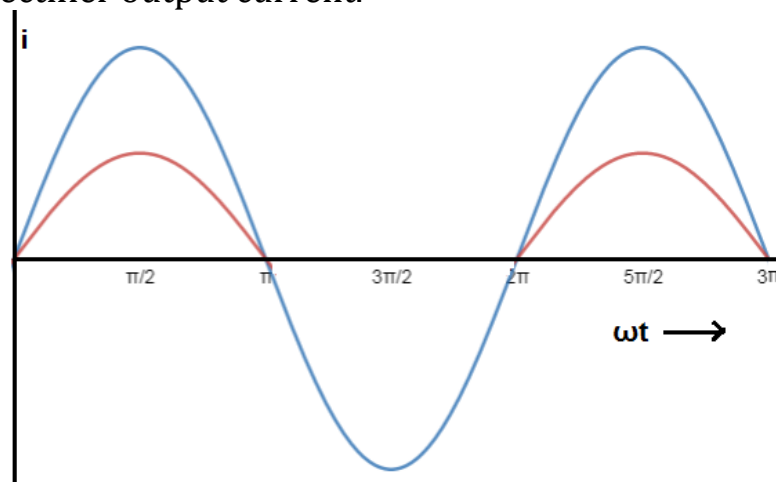
Calculation of Ripple Factor of Half Wave Rectifier

The circuit diagram of single phase half wave rectifier is shown below.



The diode will be forward biased for the positive half cycle of the supply whereas it will be reverse biased for the negative half cycle. Thus the diode will conduct only for the positive half cycle and will not conduct for the negative half cycle. Therefore, current output will only be conducting and allowing the current to flow just for the positive half cycle of the supply. This is the reason; it is called single phase half wave rectifier.

The output waveform of half wave rectifier ignoring cut-in voltage is shown in figure below. Blue color depicts the supply voltage whereas red color denotes the rectifier output current.



This output current can be defined as

$$i = I_m \sin \omega t \text{ for } 0 \leq \omega t \leq \pi$$

$$= 0 \text{ for } \pi \leq \omega t \leq 2\pi$$

Let us now find the rms value and average value of this current to find the ripple factor of half wave rectifier.

$$\begin{aligned}
RMS \text{ Value} &= \sqrt{\left(\frac{1}{2\pi}\right) \int_0^\pi (Im \sin \omega t)^2 d(\omega t) + \left(\frac{1}{2\pi}\right) \int_\pi^{2\pi} 0 d\omega t} \\
&= \sqrt{\left(\frac{Im^2}{2\pi}\right) \int_0^\pi (\sin \omega t)^2 d(\omega t)} \\
&= Im \sqrt{\left(\frac{1}{4\pi}\right) \int_0^\pi 2 (\sin \omega t)^2 d(\omega t)} \\
&= \left(\frac{Im}{2}\right) \sqrt{\left(\frac{1}{\pi}\right) \int_0^\pi (1 - \cos \omega t) d(\omega t)} \\
&= \left(\frac{Im}{2}\right) \sqrt{\left(\frac{1}{\pi}\right) \left[\int_0^\pi 1 d(\omega t) - \int_0^\pi \cos \omega t d(\omega t) \right]} \\
&= \left(\frac{Im}{2}\right) \sqrt{\left(\frac{1}{\pi}\right) [\pi - 0]} \\
&= \frac{Im}{2}
\end{aligned}$$

Thus, the rms value of current output of half wave rectifier is 0.5 times of the peak current i.e. $0.5I_m$.

The average value of single phase half wave rectifier is equal to 0.318 times of peak current i.e. $0.318I_m$. For derivation, please read “Average Value of AC Current – Definition, Formula and Application”

Now, the ripple factor of half wave rectifier can easily be calculated from the formula.

$$\begin{aligned}
\text{Ripple Factor, } \gamma &= \frac{\sqrt{(I_{rms})^2 - (I_{dc})^2}}{I_{dc}} \\
&= \frac{\sqrt{(V_{rms})^2 - (V_{dc})^2}}{V_{dc}}
\end{aligned}$$

Putting the value of rms current and average current, the value of ripple factor can easily be calculated from the above formula.

$$\text{Ripple Factor} = \frac{\sqrt{(0.5I_m)^2 - (0.318I_m)^2}}{0.318I_m}$$

$$= 1.21$$

3) A diode whose internal resistance is 20Ω is to supply power to a 100Ω load from 110V (R.M.S) source of supply. Calculate: i) Peak Load Current ii) DC Load Current iii) AC Load Current iv) % Regulation from No load to given load.

Let's assume that the supply voltage mentioned as 110V is the RMS value. Based on this assumption, we can proceed with the calculations.

Given: Internal resistance of the diode (r) = 20Ω Load resistance (R) = 100Ω AC supply voltage (V_{rms}) = 110V

i) Peak Load Current: $V_p = \sqrt{2} * V_{rms}$ $V_p = \sqrt{2} * 110\text{V}$

$I_p = V_p / (R + r)$ $I_p = (\sqrt{2} * 110\text{V}) / (100\Omega + 20\Omega)$

ii) DC Load Current: $I_{dc} = I_p - (V_p / r)$ $I_{dc} = I_p - (\sqrt{2} * 110\text{V}) / 20\Omega$

iii) AC Load Current: $I_{ac} = I_{dc} / \sqrt{2}$

iv) % Regulation from No load to given load: $V_{nl} = V_p$ $V_{fl} = V_p - (I_{dc} * r)$

% Regulation = $((V_{nl} - V_{fl}) / V_{fl}) * 100$

Now, let's calculate the values step by step:

i) Peak Load Current: $V_p = \sqrt{2} * 110\text{V}$ $V_p \approx 155.56\text{V}$

$I_p = 155.56\text{V} / (100\Omega + 20\Omega)$ $I_p \approx 1.335\text{A}$

ii) DC Load Current: $I_{dc} = 1.335\text{A} - (155.56\text{V} / 20\Omega)$ $I_{dc} \approx 1.335\text{A} - 7.778\text{A}$ $I_{dc} \approx -6.443\text{A}$ (Note: The negative sign indicates that the current direction is opposite to the assumed reference direction. The diode is reverse biased in this case, indicating no conduction.)

iii) AC Load Current: $I_{ac} = -6.443\text{A} / \sqrt{2}$ $I_{ac} \approx -4.55\text{A}$

iv) % Regulation from No load to given load: $\% \text{ Regulation} = ((V_{nl} - V_{fl}) / V_{fl}) * 100$
 $\% \text{ Regulation} = ((155.56V - (I_{dc} * 20\Omega)) / (I_{dc} * 20\Omega)) * 100$

Using the calculated values of I_{dc} from step (ii): $\% \text{ Regulation} = ((155.56V - (-6.443A * 20\Omega)) / (-6.443A * 20\Omega)) * 100$

Please note that the negative values of I_{dc} and I_{ac} indicate that the diode is reverse biased and not conducting in this scenario. The calculations show that the diode is not suitable for supplying power to the 100Ω load in this configuration.

4) Explain the operation of Full Wave Rectifier with necessary graphs.

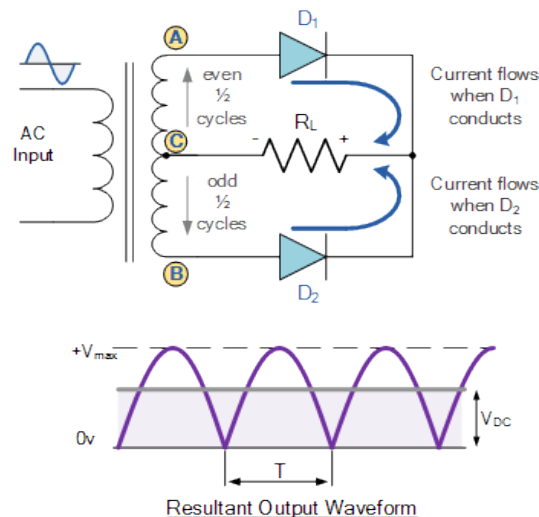
Full Wave Rectifier is a rectifier that converts both halves of each alternating wave cycle (Alternating Current) into a pulsating DC (Direct Current) signal.

- Electric circuits that are used for the conversion of AC to DC are known as rectifiers.
- There are two types of rectifiers namely Full Wave Rectifiers and Half Wave Rectifiers.
- Full-wave rectifiers are used for a more smooth and steady supply instead of a half-wave rectifier as a significant amount of power is lost while using the half-wave rectifier.
- Full-wave rectifiers are used for the conversion of a complete cycle of AC voltage to DC voltage.

What is Full Wave Rectifier

A Full Wave Rectifier is a rectifier that converts the complete cycle of an AC signal (alternating wave) into a pulsating DC signal.

- The basic difference between a half-wave rectifier and a full-wave rectifier is that the half-wave rectifier uses only the half wave of the input AC signal whereas the full-wave rectifiers use the whole cycle of the alternating wave.
- The lower efficiency of a half-wave rectifier can be overcome by using a full-wave rectifier.
- The process through which the conversion of AC to Voltage signal takes place in a full wave rectifier is known as full wave rectification.



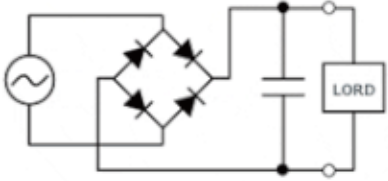
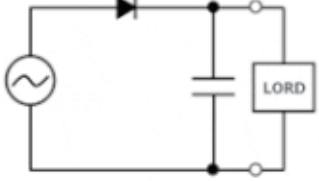
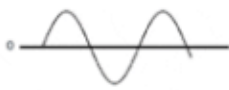
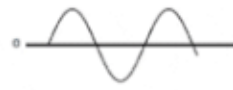
Full Wave Rectifier Circuit Diagram

Full Wave Rectifier Detailed Video Explanation:

Working Principle of Full Wave Rectifier

The Full wave rectifier follows the given working principle–

- A very high input AC is supplied to the full wave rectifier.
- The step-down transformer in the full wave rectifier circuit converts the high-voltage AC into low-voltage.
- The anode of the center-tapped diodes is connected to the secondary winding of the transformer and to the load resistor.
- During the positive half cycle of the AC, the top half of the secondary winding or terminal 1 will become positive.
- While terminal two or the second half of the winding will become negative and the center tap will be at zero potential.
- At the time of the positive half cycle, the diode D1 is forward biased and diode D2 is reverse biased. This is because it is connected to the bottom of the secondary winding.
- Hence, D1 will let the current flow, and D2 will block the flow through it.
- In the case of a negative half cycle, the diode D1 is reverse biased and the diode D2 is forward biased.
- This is because the top half of the secondary circuit becomes negative, while the bottom half of it becomes positive.
- Therefore, in a full wave rectifier, DC voltage is obtained for both positive as well as negative half cycles.

	Full-Wave Rectification	Half-Wave Rectification
Circuit Configuration		
Input Voltage Waveform		
Voltage Waveform After Rectification		
Voltage Waveform After Rectification Smoothing		

Working of Full Wave Rectifier

Applications of Full Wave Rectifier

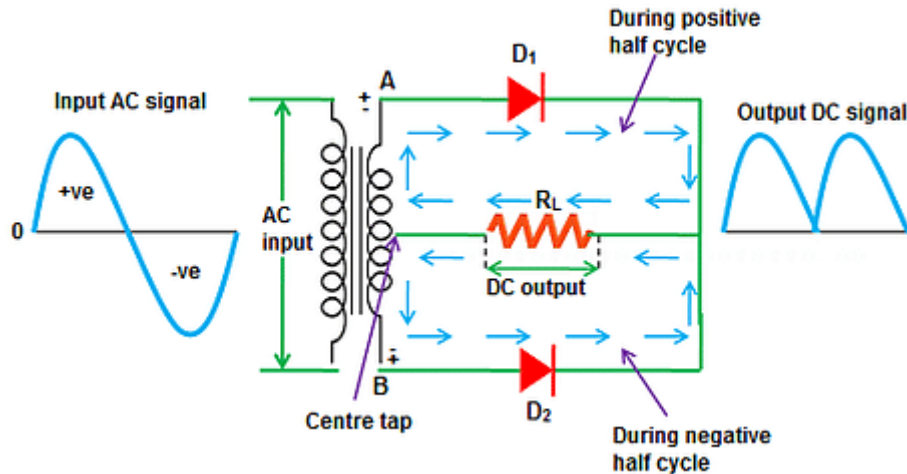
The applications of a full-wave rectifier include –

- It is used to identify the amplitude of the modulating radio signal.
- Full wave rectifiers are used in electric welding, polarized DC voltage can be supplied through a bridge rectifier.
- The bridge rectifier circuit is used as it can convert the voltage from high AC to low DC.
- They provide the power supply to the devices that function with DC voltage such as LED and Motor.

Full Wave Rectifier Circuit

A Full Wave Rectifier circuit can be made in two ways.

- The first way is by using two diodes and a center-tapped transformer, a center-tapped full-wave rectifier can be formed.
- For the second way, a standard transformer with four diodes is arranged as a bridge. This type is known as a bridge rectifier.

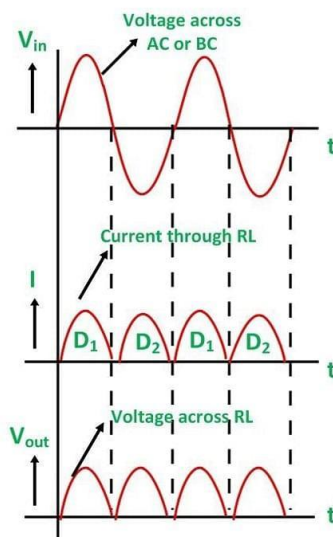


Full Wave Rectifier Diagram

Centre-tapped Full Wave Rectifier

A center-tapped full-wave rectifier system is made up of a Centre-tapped Transformer, Two Diodes, and a Resistive Load.

- In a center-tapped transformer, an additional wire is connected to the center of the second winding for tapping. It divides the second winding into two parts.
- The voltage at this center-tapped part is zero.
- This divides the entire winding into positive (V_a) and negative voltage (V_b) where the entire voltage is given by the equation, $V = V_a + V_b$



Waveform of Centre Tapped Full-wave Rectifier

From the above diagram, the output for both positive and negative half-cycles has been obtained.

5) Derive the expression for Ripple factor for Full Wave Rectifier with L-section filter.

Choke Filter

Definition: Choke filter consists of an inductor connected in series with rectifier output circuit and a capacitor connected in parallel with the load resistor. It is also called L-section filter because the inductor and capacitor are connected in the shape of inverted L. The output pulsating DC voltage from a rectifier circuit passes through the inductor or choke coil.

The inductor has low DC resistance and extremely high AC reactance. Thus, ripples get filtered through choke coil. Some of the residual ripples if present in filtered signal from inductor coil will get bypassed through the capacitor. The reason behind this is that capacitor allow AC and block DC.

Significance of Choke Filter or L-section filter

Choke filter came into existence due to shortcomings of the series inductor and shunt capacitor filter. A series inductor filter filters the output current but reduces the output current (RMS value and Peak value) up to a large extent. And the shunt capacitor filter performs filtering efficiently but increases the diode current. The excess of current in a diode may lead to its destruction.

Moreover, the ripple factor of series inductor filter is directly proportional to the load resistance it means as the load resistance increases, ripple factor also starts increasing. And in the case of shunt capacitor, the ripple factor is inversely proportional to the value of load resistance. It implies that in shunt capacitor filter the ripple factor decreases with increase in load resistance and increases with the decrease in load resistance.

Thus, for better performance, we need a filter circuit in which ripple factor is low and do not vary with the variation in load resistance. This can be achieved by using the combination of series inductor filter and shunt capacitor filter. The voltage stabilization property of shunt capacitor filter and current smoothing property of series inductor filter is utilized for the formation of choke filter or L-section filter.

The combination of series inductor filter and shunt capacitor filter is generally used for most of the applications. The combination results in two types, i.e. L-section filter and Pi filter. In this article, we will discuss the

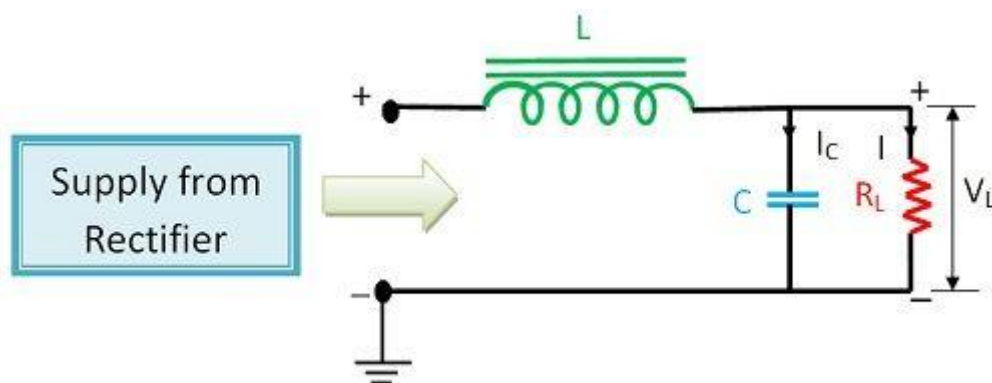
working of L-section or choke filter and in next article, we will discuss Pi filter in detail.

Working of Choke Filter or L-section filter

When the pulsating DC signal from the output of the rectifier circuit is feed into choke filter, the AC ripples present in the output DC voltage gets filtered by choke coil. The inductor has the property to block AC and pass DC. This is because DC resistance of an inductor is low and AC impedance of inductor coil is high. Thus, the AC ripples get blocked by inductor coil.

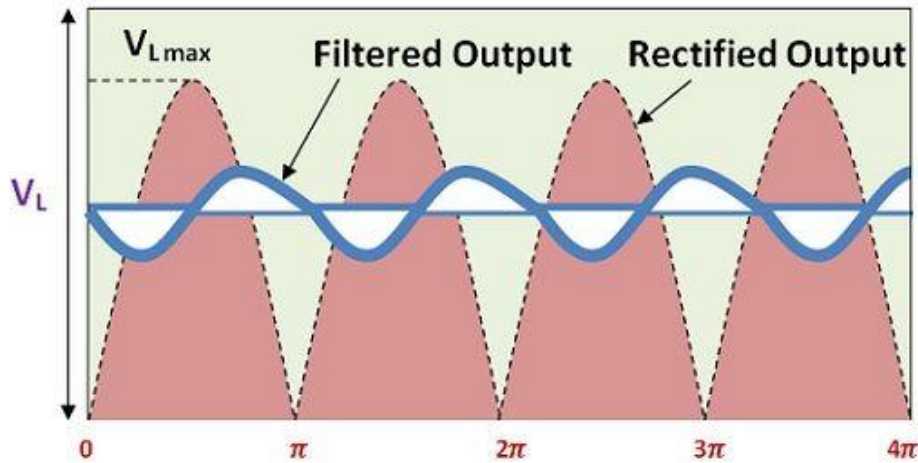
Although the inductor efficiently removes AC ripples, a small percentage of AC ripples is still present in the filtered signal. These ripples are then removed by the capacitor connected in parallel to the load resistor. Now, the DC output signal is free from AC components, and this regulated DC can be used in any application.

If the inductor of high inductive reactance (X_L), greater than the capacitive reactance at ripple frequency is used than filtering efficiency gets improved.



Waveform of Choke Filter or L-section Filter

The waveform of DC output signal with a filter and without filter is shown in the below diagram.



Characteristics of Choke filter or L-section filter

1. Regulation: The variation of DC output voltage from rectifier with respect to the DC flowing through load resistor of the rectifier circuit is termed as regulation.

$$\text{Output Voltage of Rectifier } v_L = \frac{2}{\pi} V_{L \max} - \frac{4}{3\pi} V_{L \max} \cos 2\omega t$$

Considering that the inductor's reactance is negligible in comparison to load resistance

$$V_{dc} = \frac{2}{\pi} V_{L \max}$$

If R_C is the choke resistance, then

$$V_{dc} = \frac{2}{\pi} V_{L \max} - I_{dc} R_C$$

2. Ripple factor: This is one of the significant characteristics of the filter. A filter is considered efficient if it can remove ripples effectively. The ripple factor is the ratio of the AC present in the output signal and the DC. More the ripple factor poor will be the performance of the device.

The most effective way to minimize the ripple factor is to increase the value of inductive reactance. The combined reactance of load resistor and capacitor can be minimized up to a large extent by using the capacitor of low reactance so that the complete AC signal get bypassed through the capacitor and regulated DC voltage can be obtained across the load resistor. In these conditions, the net impedance will be due to inductor coil and that will be approximately $2\omega L$ i.e. $X_L = 2\omega L$.

$$I_{ac\ rms} = \frac{4V_{L\ max}}{3\pi\sqrt{2}\ X_L} = \frac{\sqrt{2}\ V_{dc}}{3\ X_L}$$

AC Voltage across the load (the ripple voltage) is equal to voltage across the capacitor. Thus,

$$V_{ac\ rms} = I_{ac\ rms} X_c = \frac{\sqrt{2}}{3} V_{dc} \frac{X_c}{X_L}$$

Where, $X_c = \frac{1}{2\omega C}$ is the reactance of the capacitor at the second harmonic frequency.

$$\begin{aligned} \text{Ripple factor, } \gamma &= \frac{V_{ac\ rms}}{V_{dc}} \\ &= \frac{\sqrt{2}\ X_c}{3\ X_L} = \frac{\sqrt{2}}{3} \frac{1}{2\omega C} \frac{1}{2\omega L} \\ &= \frac{1}{6\omega^2 LC\sqrt{2}} \end{aligned}$$

3. Critical Inductance: The value of inductance can be increased up to a limit and this value of inductance is called critical inductance.

Advantages of Choke Filter or L-section Filter

1. It provides better voltage regulation.
2. The ripple factor can be varied according to the need.

Disadvantages of Choke Filter or L-Section filter

1. Bulky Size: These kinds of filters were popular in ancient time but it has become obsolete now due to bulky size of inductors and capacitors.
2. Not suitable for low voltage power Supplies: These are not suitable for low voltage power supplies. IC regulators or active filters are used in such devices.

It is the combination of series inductor filter and shunt capacitor filter. The advantages of both these filters are utilized to form Choke input filters. And the disadvantages of both of these filters are removed in choke filter.

6) With a neat circuit diagram, Explain the operation of Full Wave Rectifier with Capacitor filter

The device that converts Alternating Current (AC) into Direct Current (DC) is referred to as rectifier.

In half wave rectifier, the conversion of Alternating Current (AC) into Direct Current (DC) is not efficient. Half wave rectifier allows either positive half cycle or negative half cycle of the input AC signal and the remaining half cycle is blocked. As a result, a large power is wasted. Also, the output Direct Current (DC) produced by the half wave rectifier contains large ripples. This ripple voltage fluctuates with respect to time. So it is not suitable for practical applications.

To overcome these problems, we use filters at the output. Even though we use filters at the output, the DC signal obtained at the output is not a pure DC. Furthermore, the power loss is high in half wave rectifier. Therefore, to reduce the power loss and reduce the ripples at the output, we go for another type of rectifier known as full wave rectifier.

As the name suggests, the full wave rectifier rectifies both positive and negative half cycles of the input AC signal.

Even though the full wave rectifier rectify both positive and negative half cycles, the DC signal obtained at the output still contains some ripples. To reduce these ripples at the output, we use a filter.

The filter is an electronic device that converts the pulsating Direct Current into pure Direct Current.

The filter is made up of a combination of electronic components such as resistors, capacitors, and inductors. The property of inductor is that it allows the DC components and blocks the AC components. The property of a capacitor is that it allows the AC components and blocks the DC components.

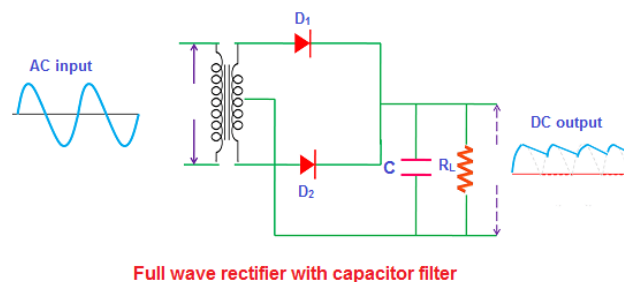
In this tutorial, a center tapped full wave rectifier with a filter made up of capacitor and resistor is explained. The filter made up of capacitor and resistor is known as capacitor filter.

In the circuit diagram, the capacitor C is placed across the load resistor R_L .

The working of the full wave rectifier with filter is almost similar to that of the half wave rectifier with filter. The only difference is that in the half wave rectifier only one half cycle (either positive or negative) of the input AC current will charge the capacitor but the remaining half cycle will not charge the capacitor. But in full wave rectifier, both positive and negative half cycles of the input AC current will charge the capacitor.

The main duty of the capacitor filter is to short the ripples to the ground and blocks the pure DC (DC components), so that it flows through the alternate path and reaches output load resistor R_L .

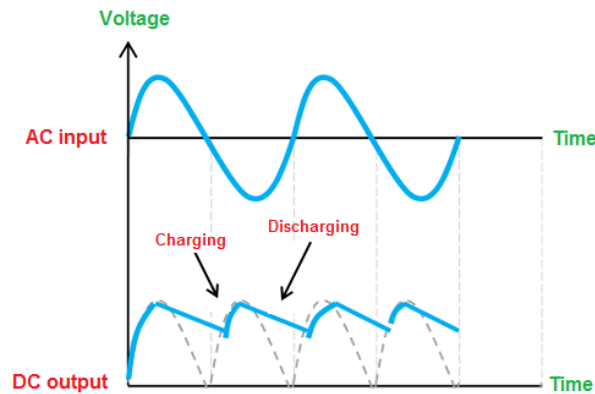
When input AC voltage is applied, during the positive half cycle, the diode D_1 is forward biased and allows electric current whereas the diode D_2 is reverse biased and blocks electric current. On the other hand, during the negative half cycle the diode D_2 is forward biased (allows electric current) and the diode D_1 is reverse biased (blocks electric current).



During the positive half cycle, the diode (D_1) current reaches the filter and charges the capacitor. However, the charging of the capacitor happens only when the applied AC voltage is greater than the capacitor voltage.

Initially, the capacitor is uncharged. That means no voltage exists between the plates of the capacitor. So when the voltage is turned on, the charging of the capacitor happens immediately.

During this conduction period, the capacitor charges to the maximum value of the input supply voltage. The capacitor stores a maximum charge exactly at the quarter positive half cycle in the waveform. At this point, the supply voltage is equal to the capacitor voltage.



When the AC voltage starts decreasing and becomes less than the capacitor voltage, then the capacitor starts slowly discharging.

The discharging of the capacitor is very slow as compared to the charging of the capacitor. So the capacitor does not get enough time to completely discharge. Before the complete discharge of the capacitor happens, the charging again takes place. So only half or more than half of the capacitor charge gets discharged.

When the input AC supply voltage reaches the negative half cycle, the diode D_1 is reverse biased (blocks electric current) whereas the diode D_2 is forward biased (allows electric current).

During the negative half cycle, the diode (D_2) current reaches the filter and charges the capacitor. However, the charging of the capacitor happens only when the applied AC voltage is greater than the capacitor voltage.

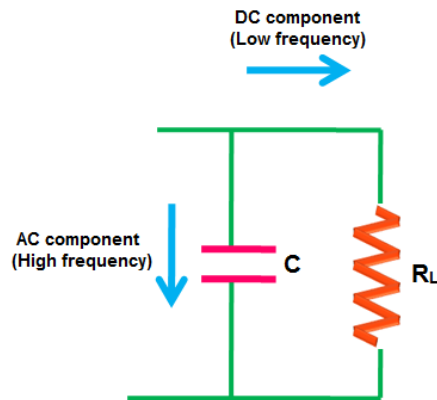
The capacitor is not completely uncharged, so the charging of the capacitor does not happen immediately. When the supply voltage becomes greater than the capacitor voltage, the capacitor again starts charging.

In both positive and negative half cycles, the current flows in the same direction across the load resistor R_L . So we get either complete positive half cycles or negative half cycles. In our case, they are complete positive half cycles.

How exactly the capacitor filter removes the ripples in the signal

The pulsating Direct Current (DC) produced by the full wave rectifier contains both AC and DC components.

We know that the capacitor allows the AC components and blocks the DC components of the current. When the DC current that contains both DC components and AC components reaches the filter, the DC components experience a high resistance from the capacitor whereas the AC components experience a low resistance from the capacitor.



Electric current always prefers to flow through a low resistance path. So the AC components will flow through the capacitor whereas the DC components are blocked by the capacitor. Therefore, they find an alternate path and reach the output load resistor R_L . The flow of AC components through the capacitor is nothing but the charging of a capacitor.

Thus, the filter converts the pulsating DC into pure DC.

7) A Half Wave Rectifier circuit supplies 100 mA DC current to a $250\ \Omega$ load. Find the DC output voltage, PIV rating of the diode and rms voltage for the transformer supplying the rectifier

Given:

- Load resistance, $R = 250\ \Omega$
- Load current, $I = 100\ \text{mA} = 0.1\ \text{A}$

Required:

- DC output voltage, V_{dc}
- Peak inverse voltage (PIV) rating of the diode
- RMS voltage for the transformer supplying the rectifier

Solution:

The DC output voltage of a half-wave rectifier is given by:

$$V_{dc} = I * R = 0.1 \text{ A} * 250 \Omega = 25 \text{ V}$$

The peak inverse voltage (PIV) rating of the diode is equal to the peak voltage of the input AC voltage. The peak voltage of the input AC voltage is given by:

$$V_{peak} = V_{dc} * \sqrt{2} = 25 \text{ V} * \sqrt{2} = 35.35 \text{ V}$$

Therefore, the PIV rating of the diode is 35.35 V.

The RMS voltage for the transformer supplying the rectifier is given by:

$$V_{rms} = V_{dc} / \sqrt{2} = 25 \text{ V} / \sqrt{2} = 17.32 \text{ V}$$

Therefore, the RMS voltage for the transformer supplying the rectifier is 17.32 V.

8) State and prove Clamping Circuit Theorem.

Clamping Circuit Theorem:

Under steady-state conditions, for any input waveform, the shape of the output waveform of a clamping circuit is fixed and also the area in the forward direction and the area in the reverse direction are related. According to clamping circuit theorem, under steady-state conditions, the ratio of area in forward direction A_f to that of reverse direction A_r of output voltage is equal to the ratio of diode forward resistance R_f to resistance R connected across diode.

$$\text{i.e., } \frac{A_f}{A_r} = \frac{R_f}{R}$$

This theorem applies quite generally independent of the input waveform and the magnitude of the source resistance.

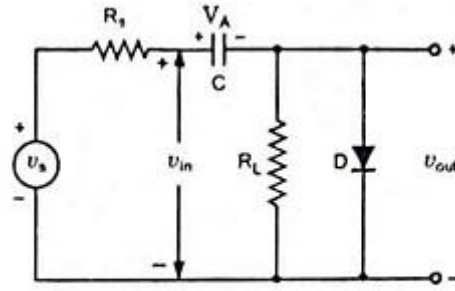


Fig. 30.79 A Practical Clamping Circuit

Proof: Consider the clamping circuit shown in Fig. 30.79. Let for first half cycle of input signal diode be on, the equivalent circuit is shown in Fig. 30.80 (a). If $v_{out f}(t)$ is the output waveform in the forward direction, then the capacitor charging current is

$$i_f(t) = \frac{v_{out f}(t)}{R_f} \quad \text{for } 0 < t < T_1$$

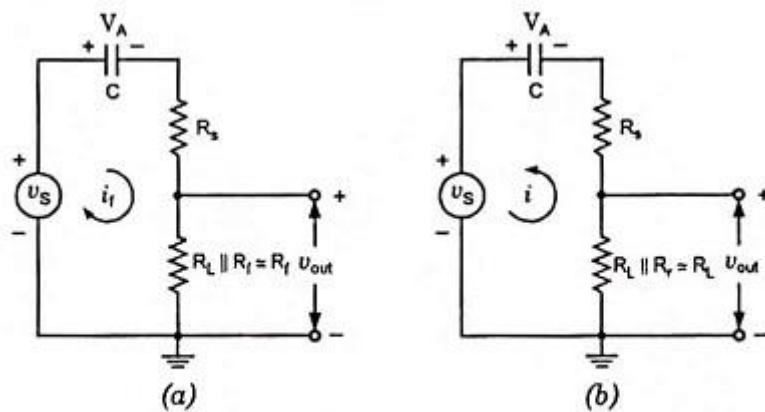


Fig. 30.80 Equivalent Circuits When Diode is Forward and Reverse Biased

Therefore, the charge stored in the capacitor during the forward interval is

$$Q_{Cf} = \int_0^{T_1} i_f(t) dt = \frac{1}{R_f} \int_0^{T_1} v_{out f}(t) dt = \frac{A_f}{R_f}$$

Similarly for interval $T_1 < t < T_1 + T_2$, the input is low and the diode is off and the equivalent circuit is as shown in Fig. 30.80 (b). If $v_{out r}(t)$ is the output voltage in the reverse direction, then the discharging current of capacitor is

$$i_r(t) = \frac{v_{out r}(t)}{R}$$

Thus, the charge lost by the capacitor during the reversal interval is

$$Q_{Cr} = \int_{T_1}^{T_1 + T_2} i_r(t) dt = \frac{1}{R} \int_{T_1}^{T_1 + T_2} v_{out,r}(t) dt = \frac{A_r}{R}$$

Under steady-state conditions, the net charge gain of the capacitor over one complete cycle must be equal to zero, therefore,

$$\begin{aligned} Q_{Cf} &= Q_{Cr} \\ \text{or } \frac{A_f}{R_f} &= \frac{A_r}{R} \\ \text{or } \frac{A_f}{A_r} &= \frac{R_f}{R} \quad \text{Hence proved.} \end{aligned}$$

9) Design a clipping circuit that clips at two independent levels.

Here is the design of a clipping circuit that clips at two independent levels.

Circuit Diagram

The circuit diagram of a clipping circuit that clips at two independent levels is shown below:

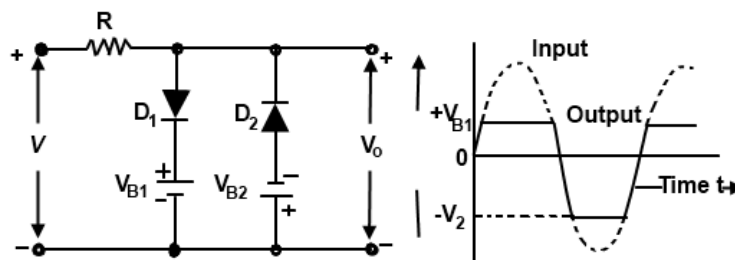


Figure 9: Dual (Combination) Diode Clipper

The circuit consists of two diodes, D1 and D2, and two resistors, R1 and R2. The diodes are connected in parallel, and the resistors are connected in series. The input signal is applied to the circuit at the input terminals.

Operation

The operation of the circuit is as follows:

- When the input signal is below the lower clipping level, both diodes are reverse biased and the output voltage is zero.
- When the input signal is between the lower and upper clipping levels, D1 is forward biased and D2 is reverse biased. The output voltage is equal to the voltage drop across R1.
- When the input signal is above the upper clipping level, both diodes are forward biased and the output voltage is equal to the voltage drop across R2.

Design Parameters

The design parameters of the circuit are as follows:

- The lower clipping level is determined by the voltage drop across R1.
- The upper clipping level is determined by the voltage drop across R2.
- The value of R1 should be chosen to be greater than the resistance of the diode.
- The value of R2 should be chosen to be greater than the resistance of the diode.

Applications

The clipping circuit can be used to clip an input signal at two independent levels. This can be useful for a variety of applications, such as:

- Limiting the amplitude of an input signal.
- Generating a square wave from a sine wave.
- Detecting the presence of a signal above or below a certain level.

10) In half-wave rectifier an ac voltage of peak value 24V is connected in series with silicon diode and load resistance of 480Ω . If the forward resistance of the diode is 20Ω , find average load current and rms value of load current

Given:

- Peak voltage of AC input, $V_m = 24\text{ V}$

- Forward resistance of diode, $R_f = 20 \Omega$
- Load resistance, $R_L = 480 \Omega$

Required:

- Average load current, I_{avg}
- RMS value of load current, I_{rms}

Solution:

The peak inverse voltage (PIV) rating of the diode is equal to the peak voltage of the input AC voltage. The PIV rating of the diode is given by:

$$PIV = V_m = 24 \text{ V}$$

The average load current is given by:

$$I_{avg} = V_m / (R_L + R_f) = 24 \text{ V} / (480 \Omega + 20 \Omega) = 45 \text{ mA}$$

The RMS value of load current is given by:

$$I_{rms} = I_{avg} * \sqrt{2} = 45 \text{ mA} * \sqrt{2} = 67.08 \text{ mA}$$

Therefore, the average load current is 45 mA and the RMS value of load current is 67.08 mA.

11) Explain the operation of negative clamping circuit.

A negative clamping circuit is a type of electronic circuit that is used to shift the negative peaks of an input signal to a lower level. The circuit is made up of a diode, a resistor, and a capacitor. The diode is connected in series with the resistor and the capacitor is connected across the diode. The input signal is applied to the circuit at the input terminals.

The operation of the negative clamping circuit is as follows:

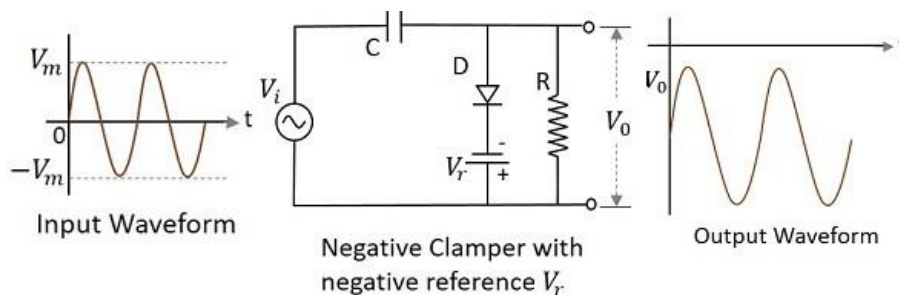
- When the input signal is negative, the diode is reverse biased and the capacitor does not charge. The output voltage is therefore equal to the negative input voltage.

- When the input signal is positive, the diode is forward biased and the capacitor starts to charge. The output voltage is therefore equal to the voltage drop across the resistor.
- When the input signal returns to negative, the capacitor discharges through the diode and the output voltage is clamped to the negative level.

The negative clamping circuit can be used to shift the negative peaks of an input signal to a lower level for a variety of purposes. For example, it can be used to:

- Protect sensitive electronic circuits from negative voltage spikes.
- Generate square waves from sine waves.
- Provide a reference voltage for other electronic circuits.

Here is a diagram of a negative clamping circuit:



The following are the components of the negative clamping circuit:

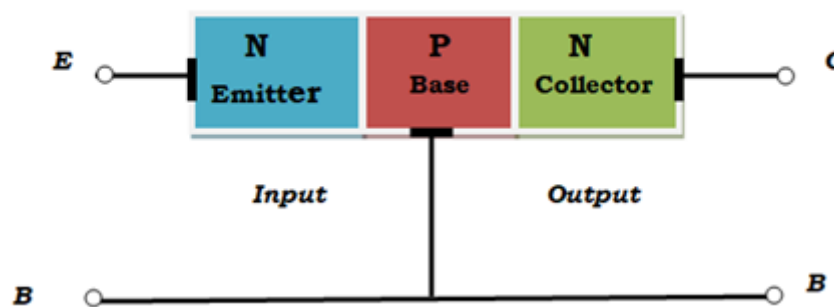
- Diode: The diode is used to block the negative input voltage from reaching the output.
- Resistor: The resistor limits the current flow through the diode and the capacitor.
- Capacitor: The capacitor stores the charge from the input signal and provides a reference voltage for the output.

UNIT 3

1.a) Explain about input and output characteristics of a transistor when it is Connected in common base configuration.

Common Base configuration, input and output characteristics

Common Base configuration:



Common Base configuration

Two terminals are needed for input and two terminals are needed for output, so one terminal is taken as common for both input and output. Based on the terminal which is taken as common there are three types of configurations. They are

- Common base configuration
- Common emitter configuration
- Common collector configuration

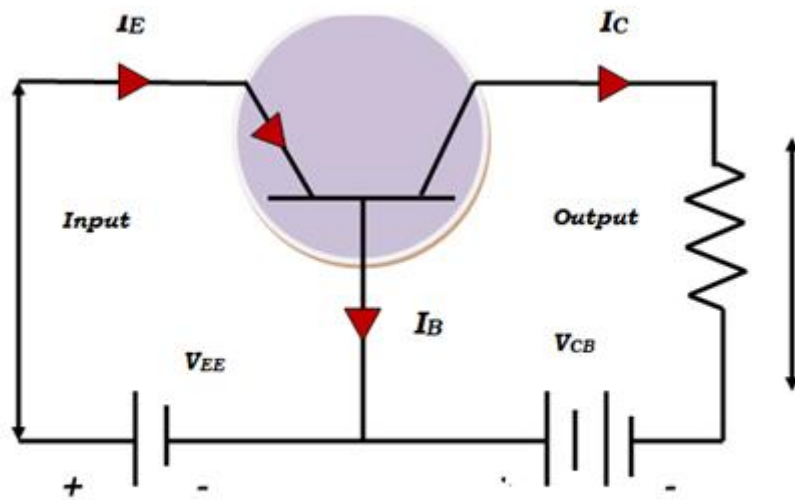
In common base configuration base terminal taken as common for both input and output. Input is applied between emitter and base terminal and output is taken between collector and base terminal.

Common emitter configuration and Common collector configuration are widely used and common base configuration is the least used.

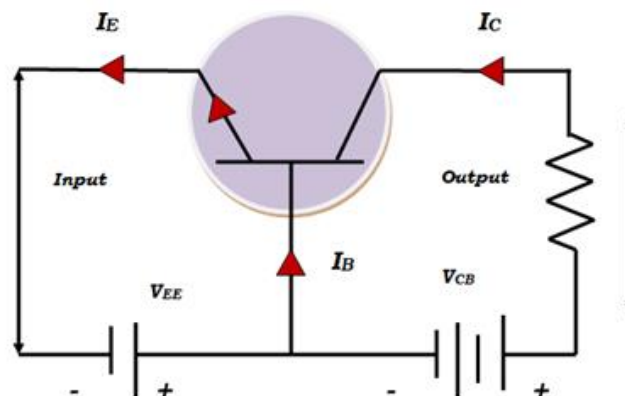
Circuit diagram of Common Base NPN and PNP transistor:

In the Common base circuit for NPN and PNP the input is given between emitter and base terminals and output is taken from collector and base terminals. The input voltage is denoted as V_{BE} and the output voltage is

denoted as V_{CE} . In all the configuration the base emitter junction is always forward biased and the collector base junction is reverse biased.



Common Base configuration of PNP transistor



Common Base configuration of NPN transistor

In the common base configuration of NPN circuit emitter is N type base is of P type and collector is of N type. The emitter base terminals are forward biased so the majority charge carriers in the emitter that is the electrons gets repelled by the negative applied voltage and in the same way the majority charge carriers in the base that is the holes gets repelled by the positive applied voltage.

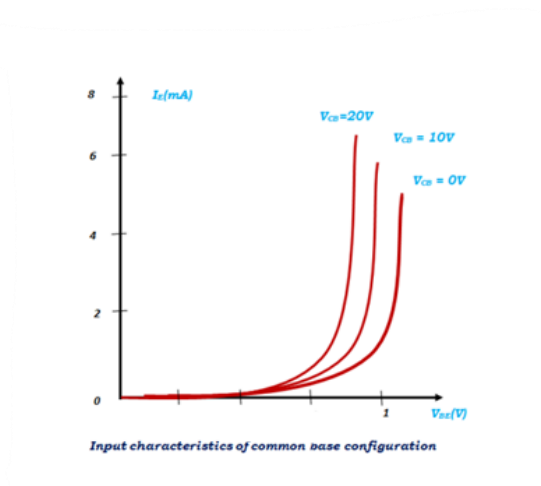
When free electrons from emitter move to the base the free electrons and the free holes combine with each other but since the base is very thin only some free electrons gets combined with the holes and the majority of the electrons are attracted towards the collector because of the positive terminal voltage connected to the collector. Thus the current flow through the output terminal.

Thus the emitter current is the sum of the base current and the collector current.

$$I_E = I_B + I_C$$

In common base configuration, the input impedance is low and the output impedance is high and the overall power gain is low when compared with other configuration.

Input characteristics:

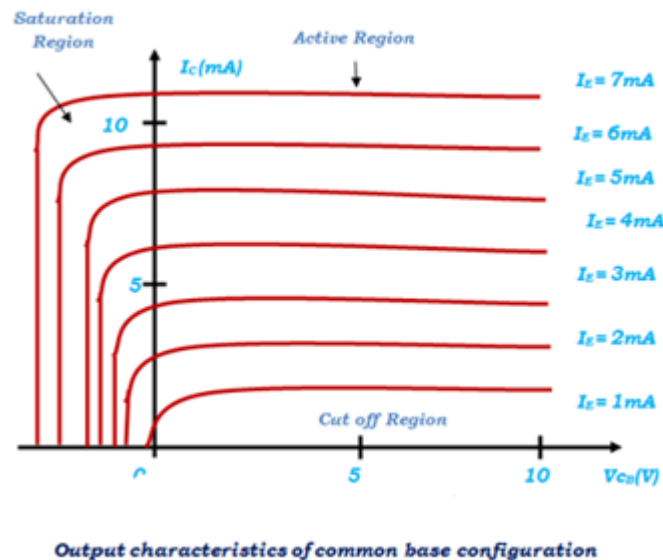


Input characteristics are the relationship between the input current and input voltage with constant output voltage. In common base configuration input current is emitter current I_E and the input voltage is base emitter voltage V_{BE} . The curve is plotted between I_E and V_{BE} keeping V_{CB} as constant.

The V_{BE} is increased keeping V_{CB} constant, initially at zero and the input current I_E is noted, similarly the V_{CB} value is increased and kept constant and V_{BE} is increased and the input current I_E is noted.

Input side is forward biased so the input resistance is small so for a small increase in V_{BE} there is rapid increase in the emitter current I_E . As the output voltage V_{CB} is increased the width of the depletion layer between emitter base decreases and the cut in voltage is reduced so the curve drifts to the left side.

Output characteristics:



Output characteristics are the relationship between output current I_C and output voltage V_{CB} keeping input current I_E constant. When the input current I_E is zero it is in cut off region. In saturation region both emitter base junction and collector base junction are forward biased.

In active region I_E is gradually increased and kept constant and output voltage V_{CB} is increased further and the output current I_C almost remains constant. So in active region curve is almost flat. Output voltage causes only a very little change in output current.

1.b) Determine I_C , I_E and α for a transistor circuit having $I_B = 15 \mu A$ and $\beta = 150$.

Given:

- Base current, $I_B = 15 \mu A = 15 \times 10^{-6} A$
- Current gain, $\beta = 150$

Required:

- Collector current, I_C
- Emitter current, I_E
- Common base current gain, α

Solution:

The collector current is given by:

$$I_C = \beta * I_B = 150 * 15 * 10^{-6} \text{ A} = 2.25 \text{ mA}$$

The emitter current is given by:

$$I_E = I_C + I_B = 2.25 \text{ mA} + 15 \mu\text{A} = 2.4 \text{ mA}$$

The common base current gain is given by:

$$\alpha = I_C / I_E = 2.25 \text{ mA} / 2.4 \text{ mA} = 0.9375$$

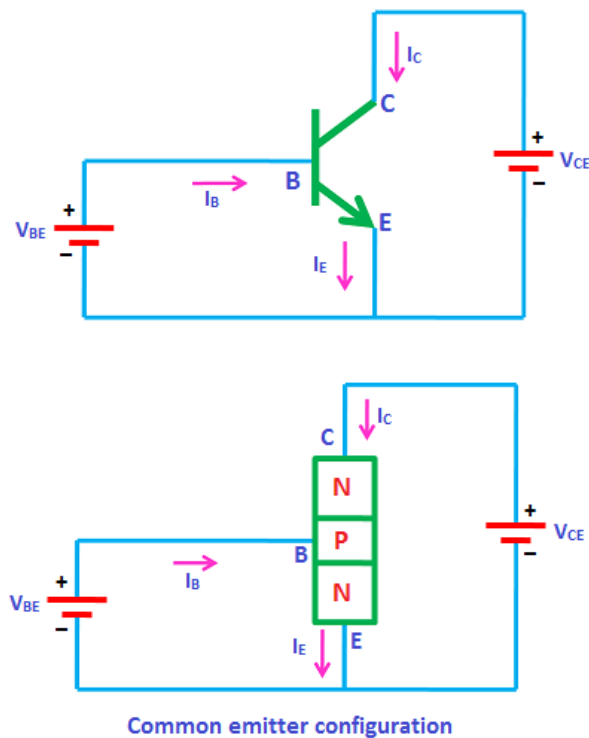
Therefore, the collector current is 2.25 mA, the emitter current is 2.4 mA, and the common base current gain is 0.9375.

2.a) Draw and explain the CE characteristics of a transistor.

Common Emitter Configuration

In common emitter configuration, base is the input terminal, collector is the output terminal and emitter is the common terminal for both input and output. That means the base terminal and common emitter terminal are known as input terminals whereas collector terminal and common emitter terminal are known as output terminals.

In common emitter configuration, the emitter terminal is grounded so the common emitter configuration is also known as grounded emitter configuration. Sometimes common emitter configuration is also referred to as CE configuration, common emitter amplifier, or CE amplifier. The common emitter (CE) configuration is the most widely used transistor configuration.



The common emitter (CE) amplifiers are used when large current gain is needed.

The input signal is applied between the base and emitter terminals while the output signal is taken between the collector and emitter terminals. Thus, the emitter terminal of a transistor is common for both input and output and hence it is named as common emitter configuration.

The supply voltage between base and emitter is denoted by V_{BE} while the supply voltage between collector and emitter is denoted by V_{CE} .

In common emitter (CE) configuration, input current or base current is denoted by I_B and output current or collector current is denoted by I_C .

The common emitter amplifier has medium input and output impedance levels. So the current gain and voltage gain of the common emitter amplifier is medium. However, the power gain is high.

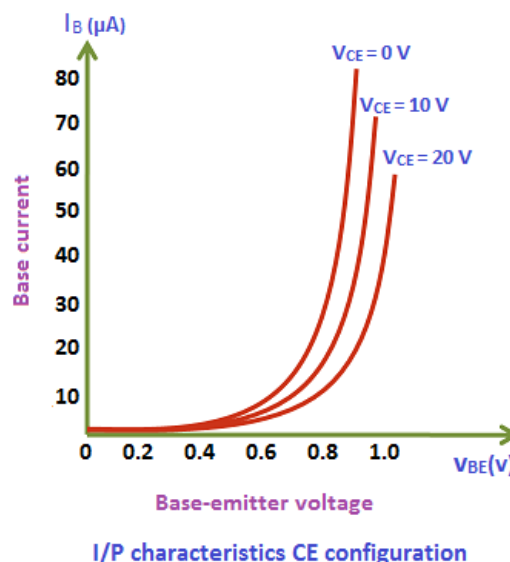
To fully describe the behavior of a transistor with CE configuration, we need two set of characteristics – input characteristics and output characteristics.

Input characteristics

The input characteristics describe the relationship between input current or base current (I_B) and input voltage or base-emitter voltage (V_{BE}).

First, draw a vertical line and a horizontal line. The vertical line represents y-axis and horizontal line represents x-axis. The input current or base current (I_B) is taken along y-axis (vertical line) and the input voltage (V_{BE}) is taken along x-axis (horizontal line).

To determine the input characteristics, the output voltage V_{CE} is kept constant at zero volts and the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded.



A curve is then drawn between input current I_B and input voltage V_{BE} at constant output voltage V_{CE} (0 volts).

Next, the output voltage (V_{CE}) is increased from zero volts to certain voltage level (10 volts) and the output voltage (V_{CE}) is kept constant at 10 volts. While increasing the output voltage (V_{CE}), the input voltage (V_{BE}) is kept constant at zero volts. After we kept the output voltage (V_{CE}) constant at 10 volts, the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded.

A curve is then drawn between input current I_B and input voltage V_{BE} at constant output voltage V_{CE} (10 volts).

This process is repeated for higher fixed values of output voltage (V_{CE}).

When output voltage (V_{CE}) is at zero volts and emitter-base junction is forward biased by input voltage (V_{BE}), the emitter-base junction acts like a normal p-n junction diode. So the input characteristics of the CE configuration is same as the characteristics of a normal pn junction diode.

The cut in voltage of a silicon transistor is 0.7 volts and germanium transistor is 0.3 volts. In our case, it is a silicon transistor. So from the above graph, we can see that after 0.7 volts, a small increase in input voltage (V_{BE}) will rapidly increases the input current (I_B).

In common emitter (CE) configuration, the input current (I_B) is very small as compared to the input current (I_E) in common base (CB) configuration. The input current in CE configuration is measured in microamperes (μA) whereas the input current in CB configuration is measured in milliamperes (mA).

In common emitter (CE) configuration, the input current (I_B) is produced in the base region which is lightly doped and has small width. So the base region produces only a small input current (I_B). On the other hand, in common base (CB) configuration, the input current (I_E) is produced in the emitter region which is heavily doped and has large width. So the emitter region produces a large input current (I_E). Therefore, the input current (I_B) produced in the common emitter (CE) configuration is small as compared to the common base (CB) configuration.

Due to forward bias, the emitter-base junction acts as a forward biased diode and due to reverse bias, the collector-base junction acts as a reverse biased diode.

Therefore, the width of the depletion region at the emitter-base junction is very small whereas the width of the depletion region at the collector-base junction is very large.

If the output voltage V_{CE} applied to the collector-base junction is further increased, the depletion region width further increases. The base region is lightly doped as compared to the collector region. So the depletion region

penetrates more into the base region and less into the collector region. As a result, the width of the base region decreases which in turn reduces the input current (I_B) produced in the base region.

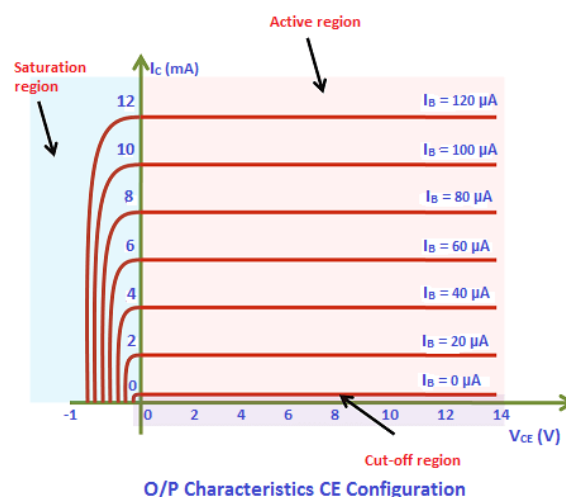
From the above characteristics, we can see that for higher fixed values of output voltage V_{CE} , the curve shifts to the right side. This is because for higher fixed values of output voltage, the cut in voltage is increased above 0.7 volts. Therefore, to overcome this cut in voltage, more input voltage V_{BE} is needed than previous case.

Output characteristics

The output characteristics describe the relationship between output current (I_C) and output voltage (V_{CE}).

First, draw a vertical line and a horizontal line. The vertical line represents y-axis and horizontal line represents x-axis. The output current or collector current (I_C) is taken along y-axis (vertical line) and the output voltage (V_{CE}) is taken along x-axis (horizontal line).

To determine the output characteristics, the input current or base current I_B is kept constant at $0\ \mu\text{A}$ and the output voltage V_{CE} is increased from zero volts to different voltage levels. For each level of output voltage, the corresponding output current (I_C) is recorded.



A curve is then drawn between output current I_C and output voltage V_{CE} at constant input current I_B ($0\ \mu\text{A}$).

When the base current or input current $I_B = 0 \mu\text{A}$, the transistor operates in the cut-off region. In this region, both junctions are reverse biased.

Next, the input current (I_B) is increased from $0 \mu\text{A}$ to $20 \mu\text{A}$ by adjusting the input voltage (V_{BE}). The input current (I_B) is kept constant at $20 \mu\text{A}$.

While increasing the input current (I_B), the output voltage (V_{CE}) is kept constant at 0 volts.

After we kept the input current (I_B) constant at $20 \mu\text{A}$, the output voltage (V_{CE}) is increased from zero volts to different voltage levels. For each voltage level of output voltage (V_{CE}), the corresponding output current (I_C) is recorded.

A curve is then drawn between output current I_C and output voltage V_{CE} at constant input current I_B ($20 \mu\text{A}$). This region is known as the active region of a transistor. In this region, emitter-base junction is forward biased and the collector-base junction is reverse biased.

This steps are repeated for higher fixed values of input current I_B (I.e. $40 \mu\text{A}$, $60 \mu\text{A}$, $80 \mu\text{A}$ and so on).

When output voltage V_{CE} is reduced to a small value (0.2 V), the collector-base junction becomes forward biased. This is because the output voltage V_{CE} has less effect on collector-base junction than input voltage V_{BE} .

As we know that the emitter-base junction is already forward biased. Therefore, when both the junctions are forward biased, the transistor operates in the saturation region. In this region, a small increase in output voltage V_{CE} will rapidly increases the output current I_C .

Transistor parameters

Dynamic input resistance (r_i)

Dynamic input resistance is defined as the ratio of change in input voltage or base voltage (V_{BE}) to the corresponding change in input current or base current (I_B), with the output voltage or collector voltage (V_{CE}) kept at constant.

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B}, \quad V_{CE} = \text{constant}$$

In CE configuration, the input resistance is very low.

Dynamic output resistance (r_o)

Dynamic output resistance is defined as the ratio of change in output voltage or collector voltage (V_{CE}) to the corresponding change in output current or collector current (I_C), with the input current or base current (I_B) kept at constant.

$$r_o = \frac{\Delta V_{CE}}{\Delta I_C}, \quad I_B = \text{constant}$$

In CE configuration, the output resistance is high.

Current gain (α)

The current gain of a transistor in CE configuration is defined as the ratio of output current or collector current (I_C) to the input current or base current (I_B).

$$\alpha = \frac{I_C}{I_B}$$

The current gain of a transistor in CE configuration is high. Therefore, the transistor in CE configuration is used for amplifying the current.

b) The reverse leakage current of the transistor when in CB configuration is $0.3\mu\text{A}$ while it is $16\mu\text{A}$ when the same transistor is connected in CE configuration. Determine α , β and γ .

Given:

- Reverse leakage current in CB configuration, $I_{CBO} = 0.3\mu\text{A}$
- Reverse leakage current in CE configuration, $I_{CEO} = 16\mu\text{A}$

Required:

- Common base current gain, α
- Common emitter current gain, β
- Collector-base current gain, γ

Solution:

The common base current gain, α is given by:

$$\alpha = I_{CEO} / I_{CBO} = 16 \mu A / 0.3 \mu A = 53.33$$

The common emitter current gain, β is given by:

$$\beta = \alpha * (1 + I_{CBO} / I_E) = 53.33 * (1 + 0.3 \mu A / 2.4 \text{ mA}) = 150$$

The collector-base current gain, γ is given by:

$$\gamma = 1 + \alpha = 1 + 53.33 = 54.33$$

Therefore, the common base current gain is 53.33, the common emitter current gain is 150, and the collector-base current gain is 54.33.

3) Explain the working of the transistor as a switch.

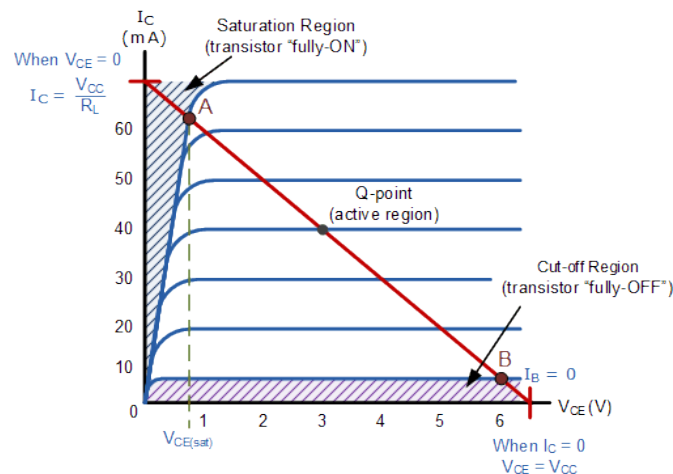
When used as an AC signal amplifier, the transistors Base biasing voltage is applied in such a way that it always operates within its “active” region, that is the linear part of the output characteristics curves are used. However, both the NPN & PNP type bipolar transistors can be made to operate as “ON/OFF” type solid state switch by biasing the transistors Base terminal differently operating the transistor as a switch.

Solid state switches are one of the main applications for the use of transistor to switch a DC output “ON” or “OFF”. Some output devices, such as LED’s only require a few milliamps at logic level DC voltages and can therefore be driven directly by the output of a logic gate. However, high power devices such as motors, solenoids or lamps, often require more power than that supplied by an ordinary logic gate so transistor switches are used.

If the circuit uses the Bipolar Transistor as a Switch, then the biasing of the transistor, either NPN or PNP is arranged to operate the transistor at both sides of the “ I-V ” characteristics curves we have seen previously.

The areas of operation for a transistor switch are known as the Saturation Region and the Cut-off Region. This means then that we can ignore the operating Q-point biasing and voltage divider circuitry required for amplification, and use the transistor as a switch by driving it back and forth between its “fully-OFF” (cut-off) and “fully-ON” (saturation) regions as shown below.

Operating Regions



The pink shaded area at the bottom of the curves represents the “Cut-off” region while the blue area to the left represents the “Saturation” region of the transistor. Both these transistor regions are defined as:

1. Cut-off Region

Here the operating conditions of the transistor are zero input base current (I_B), zero output collector current (I_C) and maximum collector voltage (V_{CE}) which results in a large depletion layer and no current flowing through the device. Therefore the transistor is switched “Fully-OFF”.

Cut-off Characteristics

The diagram shows a common-emitter transistor circuit. The base is connected to an input voltage V_{in} through a resistor R_{in}. The collector is connected to a load resistor R_L and a supply voltage V_{CC}. The emitter is grounded (0V). The output voltage V_{out} is taken from the collector. The transistor is shown in the cut-off region, with the base-emitter junction reverse-biased and the base-collector junction reverse-biased. The output is labeled 'Switch "open"'. The collector current I_C is zero, and the collector voltage V_{CE} is equal to V_{CC}.	• The input and Base are grounded (0v) • Base-Emitter voltage $V_{BE} < 0.7v$ • Base-Emitter junction is reverse biased • Base-Collector junction is reverse biased • Transistor is “fully-OFF” (Cut-off region)
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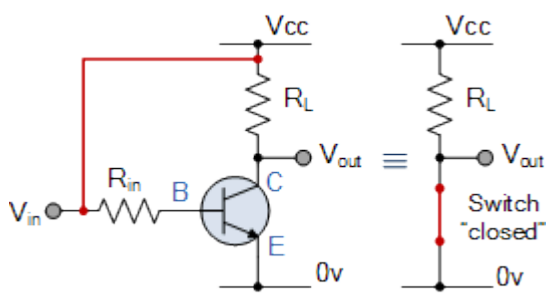
	• No Collector current flows ($I_C = 0$) • $V_{OUT} = V_{CE} = V_{CC} = "1"$ • Transistor operates as an "open switch"
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Then we can define the "cut-off region" or "OFF mode" when using a bipolar transistor as a switch as being, both junctions reverse biased, $V_B < 0.7\text{v}$ and $I_C = 0$. For a PNP transistor, the Emitter potential must be negative with respect to the Base.

2. Saturation Region

Here the transistor will be biased so that the maximum amount of base current is applied, resulting in maximum collector current resulting in the minimum collector emitter voltage drop which results in the depletion layer being as small as possible and maximum current flowing through the transistor. Therefore the transistor is switched "Fully-ON".

Saturation Characteristics

	• The input and Base are connected to V_{CC} • Base-Emitter voltage $V_{BE} > 0.7\text{v}$ • Base-Emitter junction is forward biased • Base-Collector junction is forward biased • Transistor is "fully-ON" (saturation region) • Max Collector current flows ($I_C = V_{CC}/R_L$) • $V_{CE} = 0$ (ideal saturation) • $V_{OUT} = V_{CE} = "0"$ • Transistor operates as a "closed switch"
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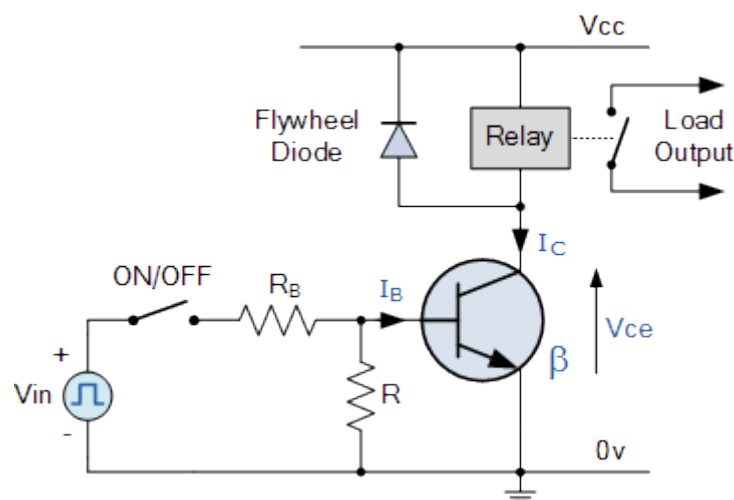
Then we can define the “saturation region” or “ON mode” when using a bipolar transistor as a switch as being, both junctions forward biased, $V_B > 0.7\text{V}$ and $I_C = \text{Maximum}$. For a PNP transistor, the Emitter potential must be positive with respect to the Base.

Then the transistor operates as a “single-pole single-throw” (SPST) solid state switch. With a zero signal applied to the Base of the transistor it turns “OFF” acting like an open switch and zero collector current flows. With a positive signal applied to the Base of the transistor it turns “ON” acting like a closed switch and maximum circuit current flows through the device.

The simplest way to switch moderate to high amounts of power is to use the transistor with an open-collector output and the transistors Emitter terminal connected directly to ground. When used in this way, the transistors open collector output can thus “sink” an externally supplied voltage to ground thereby controlling any connected load.

An example of an NPN Transistor as a switch being used to operate a relay is given below. With inductive loads such as relays or solenoids a flywheel diode is placed across the load to dissipate the back EMF generated by the inductive load when the transistor switches “OFF” and so protect the transistor from damage. If the load is of a very high current or voltage nature, such as motors, heaters etc, then the load current can be controlled via a suitable relay as shown.

Basic NPN Transistor Switching Circuit



The circuit resembles that of the Common Emitter circuit we looked at in the previous tutorials. The difference this time is that to operate the transistor as a switch the transistor needs to be turned either fully “OFF” (cut-off) or fully “ON” (saturated).

An ideal transistor switch would have infinite circuit resistance between the Collector and Emitter when turned “fully-OFF” resulting in zero current flowing through it and zero resistance between the Collector and Emitter when turned “fully-ON”, resulting in maximum current flow.

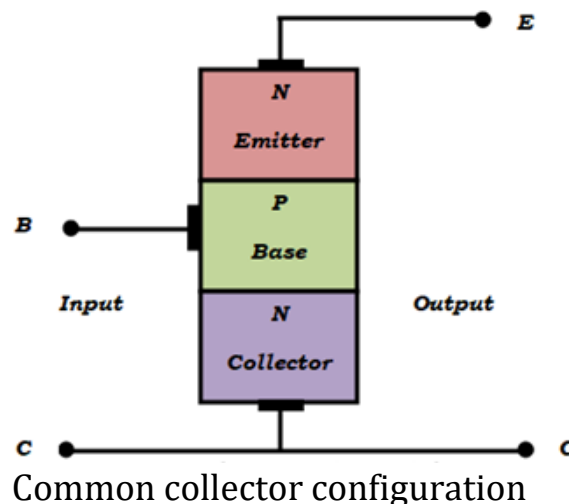
In practice when the transistor is turned “OFF”, small leakage currents flow through the transistor and when fully “ON” the device has a low resistance value causing a small saturation voltage (V_{CE}) across it. Even though the transistor is not a perfect switch, in both the cut-off and saturation regions the power dissipated by the transistor is at its minimum.

In order for the Base current to flow, the Base input terminal must be made more positive than the Emitter by increasing it above the 0.7 volts needed for a silicon device. By varying this Base-Emitter voltage V_{BE} , the Base current is also altered and which in turn controls the amount of Collector current flowing through the transistor as previously discussed.

When maximum Collector current flows the transistor is said to be Saturated. The value of the Base resistor determines how much input voltage is required and corresponding Base current to switch the transistor fully “ON”.

4.a) Explain input and output characteristics of a transistor in CC configuration with neat sketches.

Common Collector Configuration:



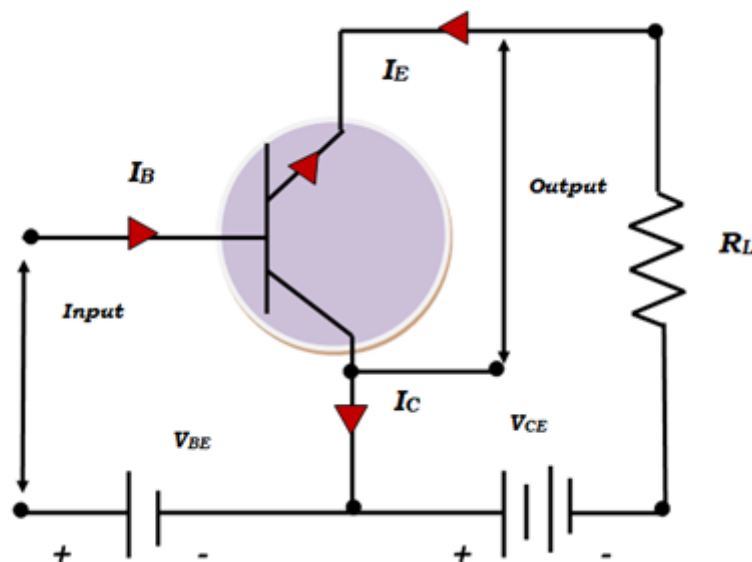
Two terminals are needed for input and two terminals for output. Transistors have three terminals, so one terminal have to be taken as common terminal for both input and output.

In common collector configuration collector terminal is taken as common. So input is applied between base and the collector terminals and output is taken from emitter and collector terminals.

The common collector configuration is also called emitter follower or voltage follower because the output emitter voltage always follows the base input voltage.

For example the base emitter voltage is 0.7v and if the input is 5V then the output is 4.3V. Output voltage is always near the input voltage. This configuration is widely used as a buffer and it is also called as voltage buffer.

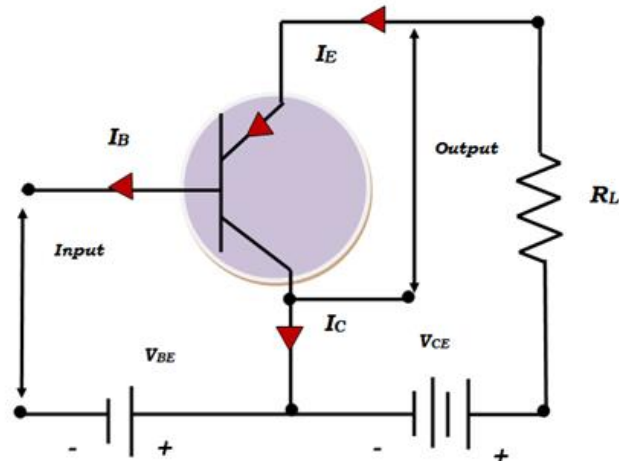
Circuit diagram of NPN and PNP configuration:



Common collector configuration of NPN transistor

Input is applied between the base and the collector terminals. Input current that is the base current is denoted as I_B and the input voltage that is the base emitter voltage is denoted as V_{BE} . Collector terminal which is taken as common is grounded.

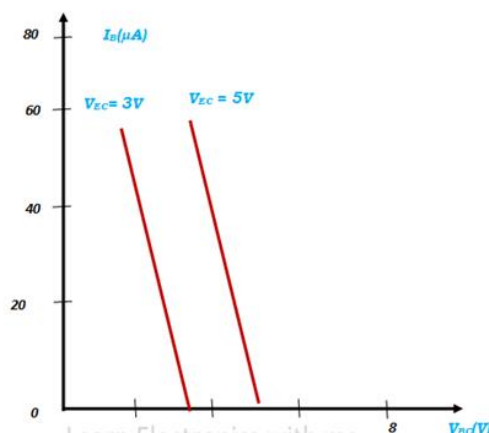
The output current is the emitter current, it is denoted as I_E and the output voltage is emitter collector voltage and it is denoted as V_{CE} . The output emitter current is the summation of the base and collector current. Emitter junction is forward biased and collector junction is reverse biased.



Common collector configuration of PNP transistor

Common collector configuration have high input configuration and low output impedance. It has low voltage gain, high current gain and the power gain is medium. This configuration is mostly used for impedance matching, that is high impedance source is used to drive low impedance load.

Input Characteristics of Common Collector Configuration:



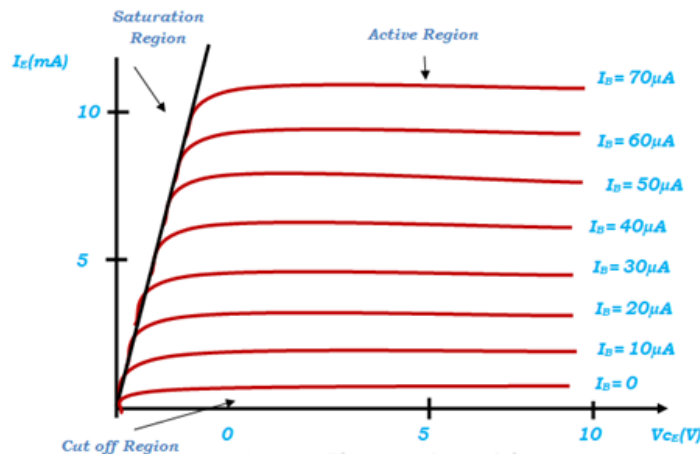
Input characteristics of common collector configuration

Input characteristics are the relationship between the input current and input voltage keeping output voltage constant. Here input current is I_B and input voltage V_{BE} and output voltage is V_{CE} .

The output voltage V_{CE} initially kept at 3V and kept constant. The input voltage V_{BE} is increased from zero gradually and the corresponding input current I_B is noted.

Then the output voltage which is kept constant is increased, let's say 5V and kept constant and output voltage is gradually increased and input current is noted. Graph is drawn with all the values noted.

Output Characteristics of Common Collector Configuration:



Output characteristics of common collector configuration

Output characteristics are the relationship between the output current and output voltage keeping input current constant. Here output current is I_E and output voltage is V_{CE} and the input current is I_B .

Initially the input current I_B is kept at zero and kept constant. Slowly input current I_B is increased like 10 μA , 20 μA and kept constant and the output voltage V_{CE} is increased gradually from zero and the corresponding output current I_E is noted.

When the input current is zero no current flows in the transistor and it is called cut off region. When the input current is very high the current through the transistor is also very high and the transistor will be in saturation region.

The region where there is a change in the output current for the change in output voltage is the active region. Here the active region almost looks flat.

4. b) A transistor has a typical β of 100. If the collector current is 40 mA, what is the value of emitter current?

The emitter current is the sum of the collector current and the base current. Given that the β of the transistor is 100 and the collector current is 40 mA, the emitter current is:

$$\begin{aligned} I_e &= I_c + I_b \\ &= 40 \text{ mA} + I_b \\ &= 40 \text{ mA} + (1/100) * 40 \text{ mA} \\ &= 40 \text{ mA} + 0.4 \text{ mA} \\ &= 40.4 \text{ mA} \end{aligned}$$

Therefore, the value of the emitter current is 40.4 mA.

5.a) Determine I_C , I_E and α for a transistor circuit having $I_B=15\mu\text{A}$ and $\beta=150$.

Given that $I_B = 15 \mu\text{A}$ and $\beta = 150$, the collector current is:

$$I_C = \beta * I_B = 150 * 15 \mu\text{A} = 2.25 \text{ mA}$$

The emitter current is:

$$I_E = I_C + I_B = 2.25 \text{ mA} + 15 \mu\text{A} = 2.4 \text{ mA}$$

The common-base current gain α is:

$$\alpha = I_C / I_E = 2.25 \text{ mA} / 2.4 \text{ mA} = 0.9375$$

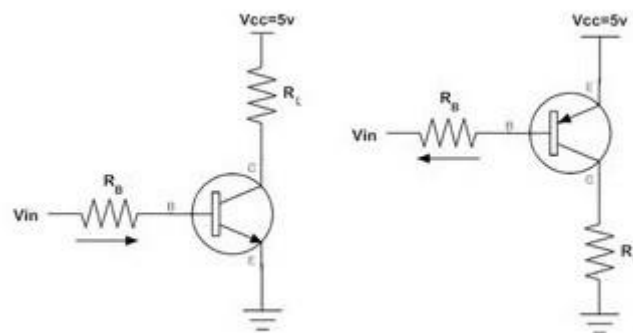
Therefore, the values of I_C , I_E , and α are 2.25 mA, 2.4 mA, and 0.9375, respectively.

5.b) With necessary circuit and waveform, explain the switching characteristics of a transistor in detail.

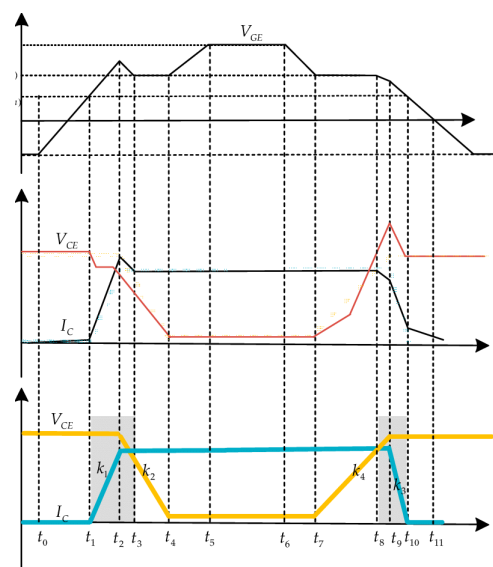
The switching characteristics of a transistor are the parameters that describe how the transistor switches between its on and off states. These parameters include the turn-on time, the turn-off time, and the saturation voltage.

The turn-on time is the time it takes for the transistor to go from the cut-off region to the saturation region. The turn-off time is the time it takes for the transistor to go from the saturation region to the cut-off region. The saturation voltage is the voltage drop across the collector-emitter junction when the transistor is in the saturation region.

The following circuit shows the switching characteristics of a transistor:



The transistor is connected to a resistor and a load. The base current is applied to the base of the transistor through a switch. When the switch is closed, the base current flows through the transistor, and the transistor turns on. The collector current flows through the collector-emitter junction, and the load is turned on. The waveform below shows the collector current waveform of the transistor:



The turn-on time is the time it takes for the collector current to rise from zero to its maximum value. The turn-off time is the time it takes for the collector current to fall from its maximum value to zero. The saturation voltage is the voltage drop across the collector-emitter junction when the collector current is at its maximum value.

The switching characteristics of a transistor are important in many applications. For example, they are important in switching power supplies, where the transistor is used to switch the power on and off very quickly. They are also important in digital circuits, where the transistor is used to switch between logic levels.

6) Explain principal of operation of BJT

A Bipolar Junction Transistor (BJT) is a three-terminal device which consists of two pn-junctions formed by sandwiching either p-type or n-type semiconductor material between a pair of opposite type semiconductors.

The primary function of BJT is to increase the strength of a weak signal, i.e., it acts as an amplifier. A BJT can also be used as a solid state switch in electronic circuits.

Types of BJT

There are two types of BJTs –

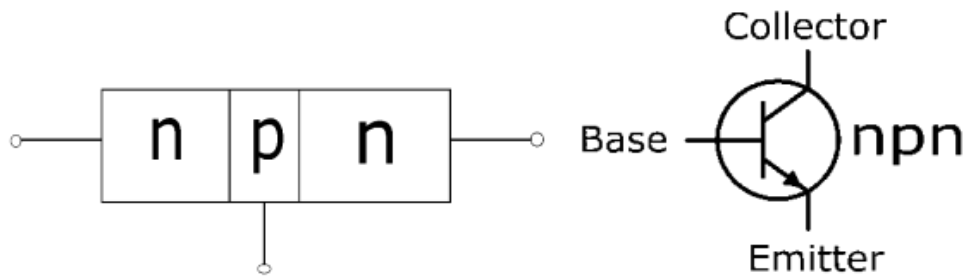
- NPN Transistor
- PNP Transistor

In this article, we will discuss in detail the working principle of both these types of BJTs.

NPN Transistor

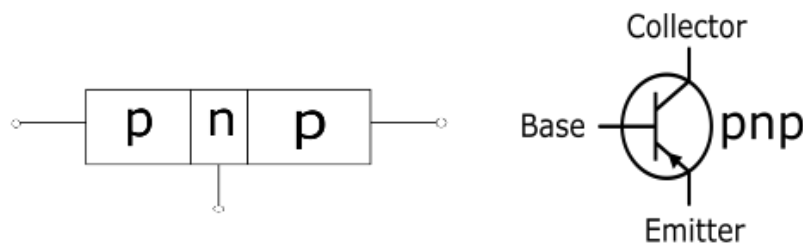
An npn-transistor is composed of two n-type semiconductor materials which are separated by a thin layer of p-type semiconductor. The two terminals viz. Emitter and Collector are taken out from the two n-type semiconductor and the Base terminal is from the p-type semiconductor.

In BJT symbol, the arrow on the emitter terminal indicates the direction of conventional current in the emitter with forward bias. For npn-transistor, the conventional current flows out of the emitter as indicated by the outgoing arrow.



PNP Transistor

A pnp-transistor is composed of two p-type semiconductors which are separated by a thin layer of n-type material. The two terminals viz. Emitter and Collector are taken out from the two p-type semiconductor layers and the Base terminal is from the n-type semiconductor. For the pnp-transistor, the conventional current flows into the emitter as indicated by the inward arrow.



Important Facts about BJT

- There are two pn-junctions, hence a transistor may be regarded as a combination of two back-to-back connected diodes.
- The collector region is wider than both emitter and base. The base is much thinner than both emitter and collector. During the transistor operation, a lots of heat is generated at the collector, hence the collector is made larger to dissipate the heat.
- A transistor has three sections of doped semiconductors. The one section is called the Emitter, the other is called the Collector, and the middle section is called the Base and forms two pn-junctions between emitter and collector.
- In general, the emitter-base junction of the BJT is made forward-biased, whereas the collector-base junction is reverse-biased.
- The resistance of forward-biased junction is very small as compared to that of the reverse-biased junction.
- The emitter is heavily doped so that it can supply a greater number of charge carriers (electrons or holes) to the base. The base is lightly doped and very thin, hence it passes most of the

charge carriers injected by the emitter to the collector. The doping concentration of the collector region is moderate.

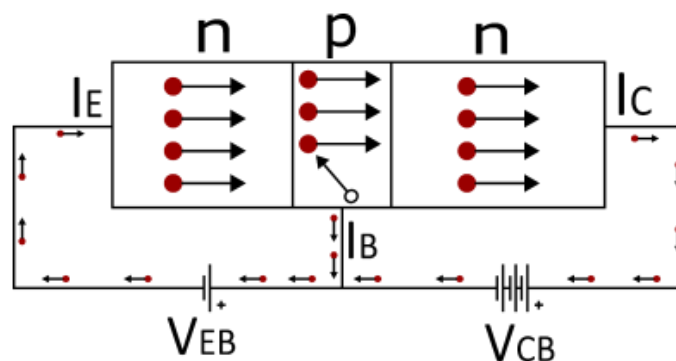
Working Principle of BJT

The emitter-base junction of BJT is forward-biased, whereas the collector-base junction is reverse biased. The forward bias of the emitter-base junction causes the emitter current to flow and this emitter current entirely flows in the collector circuit. Therefore, the collector current depends upon the emitter current and nearly equal to the emitter current.

Working of NPN Transistor

With the forward-biased emitter-base junction and reverse-biased collector-base junction, it can be seen that the forward bias causes the flow of electrons from the n-type emitter into the p-type base. This constitutes the emitter current (I_E). As these electrons flow through the p-type base, they tend to combine with the holes.

Since the base is lightly doped and very thin, hence, only a small number electrons (less than 5%) combine with the holes to constitute the base current (I_B). The remaining (more than 95%) electrons cross over the base region and reach to the collector region to constitute the collector current (I_C). In this manner, the entire emitter current flows in the collector circuit.



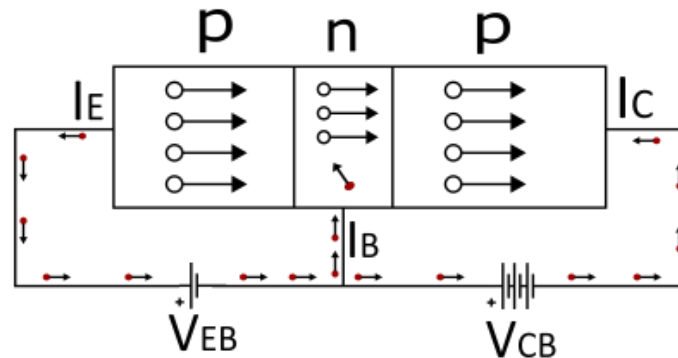
The emitter current is the sum of base and collector currents.

$$I_E = I_B + I_C$$

Working of PNP Transistor

For the pnp-transistor, the forward bias of emitter-base junction causes the flow of holes in the p-type emitter region towards the n-type base and constitutes the emitter current (I_E). As these holes cross into the n-type base region, they tend to combine with the electrons. Since the base is lightly doped and very thin, hence only a small number of holes (less than 5%) combine with the electrons. The remaining (more than 95%) cross the base and reach into the collector region to constitute the collector current (I_C).

In this manner, the entire emitter current flows into the collector circuit. It may be noted that the current conduction inside the pnp-transistor is due to the movement of holes. However, in the external connecting wires, the current is still due to the flow of electrons.

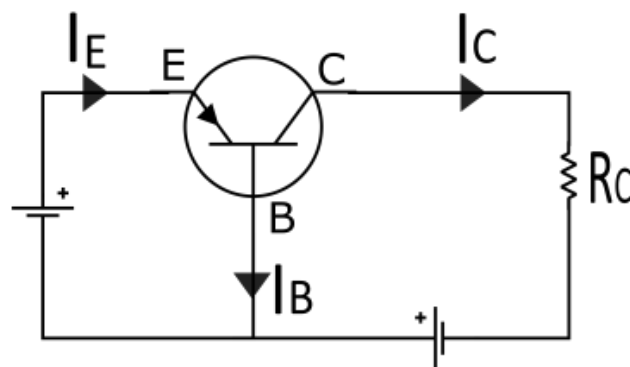


Again, the emitter current is the sum of collector current and base current.

$$I_E = I_B + I_C$$

BJT Biasing

A BJT has two pn-junctions *viz.* emitter-base junction and collector-base junction. Application of proper DC voltage at the two junctions of the BJT is known as BJT or Transistor Biasing.



When a transistor used as an amplifier, the emitter-base junction is forward biased and collector-base junction is reverse biased. If the transistor is operated under this bias condition then it is said to be operating in the active region.

When both the junctions are forward biased then the transistor is said to be operating in the saturation region. The transistor operated in saturation region acts like a closed switch and the collector current becomes maximum.

When both the junctions are reverse biased, the transistor is said to be operating in the cut off region. The BJT operated in cut off region acts as an

open switch and a very small collector current (in μA) flows from emitter to collector. This current is called reverse leakage current and is due to minority charge carriers (electrons in p-region and holes in n-region).

Emitter-Base Junction	Collector-Base Junction	Operating Region
Forward Biased	Reverse Biased	Active Region
Forward Biased	Forward Biased	Saturation Region
Reverse Biased	Reverse Biased	Reverse Biased

UNIT4

1) Explain the construction of JFET and write down principle of operation of JFET?

Here is the construction and principle of operation of a JFET, with neat diagrams:

Construction of JFET

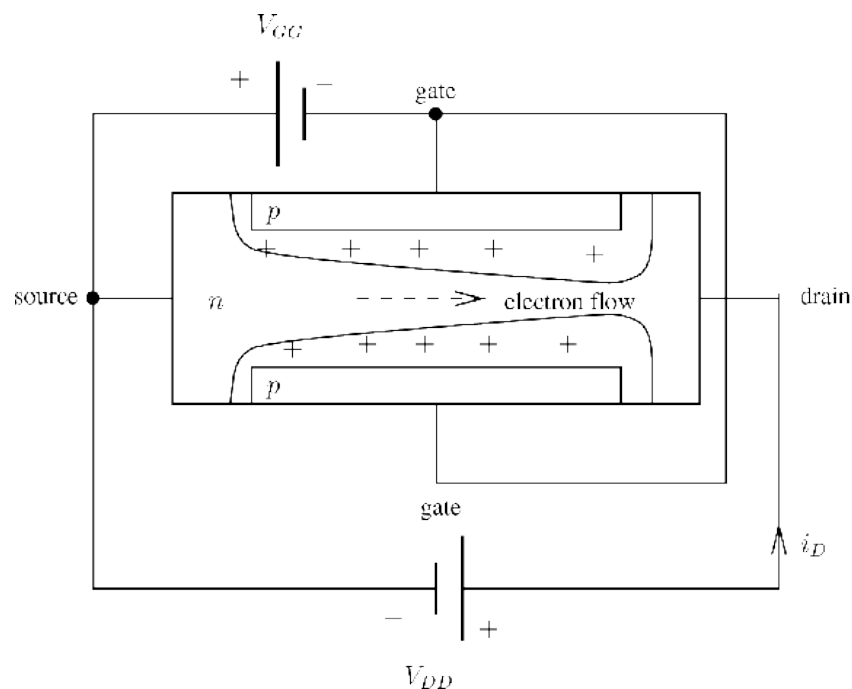
A JFET (Junction Field Effect Transistor) is a three-terminal semiconductor device. It is made up of a long, narrow channel of semiconductor material, with two ohmic contacts at the ends of the channel to form the source and drain terminals. The gate terminal is a third contact that is placed in the middle of the channel, and it is insulated from the channel by a thin layer of oxide.

The channel material is either N-type or P-type, and the gate material is the opposite type. For example, if the channel material is N-type, then the gate material is P-type.

Principle of Operation

The principle of operation of a JFET is based on the electric field that is created by the gate voltage. When a positive voltage is applied to the gate, it creates an electric field that attracts electrons to the gate. This creates a

depletion region in the channel, which is a region that is depleted of charge carriers.



The depletion region acts as a barrier to the flow of current through the channel. The larger the gate voltage, the wider the depletion region, and the smaller the current that can flow through the channel.

The drain current is controlled by the gate voltage. When the gate voltage is zero, the depletion region is very narrow, and the maximum current can flow through the channel. As the gate voltage is increased, the depletion region widens, and the current through the channel decreases.

The JFET can be operated in three different regions:

- Cut-off region: This is the region where the gate voltage is so high that the depletion region completely blocks the flow of current through the channel. [Diagram of a JFET in the cut-off region]
- Ohmic region: This is the region where the gate voltage is low enough to allow some current to flow through the channel. The current through the channel is proportional to the gate voltage. [Diagram of a JFET in the ohmic region]
- Saturation region: This is the region where the gate voltage is so high that the depletion region is so wide that the channel is almost

completely pinched off. The current through the channel is almost constant, regardless of the gate voltage. [Diagram of a JFET in the saturation region]

JFETs are used in a variety of applications, including amplifiers, oscillators, and switches. They are a versatile type of transistor that is well-suited for applications where high input impedance and low noise are required.

2) Why we call FET as a Voltage Controlled Device

FETs are called voltage-controlled devices because the current flow between the drain and source terminals is controlled by the voltage applied to the gate terminal. The gate terminal is insulated from the channel, so no current flows through it. The voltage applied to the gate creates an electric field that affects the flow of charge carriers in the channel.

In a JFET, the electric field created by the gate voltage attracts or repels charge carriers in the channel, depending on the type of FET. This changes the width of the depletion region, which in turn changes the current flow between the drain and source terminals.

In a MOSFET, the electric field created by the gate voltage causes the formation of a thin layer of charge carriers at the interface between the gate oxide and the semiconductor channel. This layer of charge carriers acts like a conducting channel, which allows current to flow between the drain and source terminals.

The voltage-controlled nature of FETs makes them very versatile devices. They can be used as amplifiers, oscillators, switches, and other electronic components.

Here are some of the advantages of using FETs as voltage-controlled devices:

- They have a high input impedance, which means that they do not draw much current from the source.
- They have a low noise level, which makes them well-suited for applications where noise is a concern.

- They can be operated at high frequencies, which makes them suitable for applications such as radio and telecommunications.

Overall, FETs are a versatile and powerful type of transistor that can be used in a wide variety of applications. Their voltage-controlled nature makes them particularly well-suited for applications where high input impedance, low noise, and high frequency operation are required.

3.a) A FET follows the relation $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$. What are the values of I_D and g_m for $V_{GS} = -1.5$ V if I_{DSS} and V_P are given as 8.4 mA and -3 V respectively?

Given that $I_{DSS} = 8.4$ mA and $V_P = -3$ V, the drain current I_D for $V_{GS} = -1.5$ V is:

$$\begin{aligned} I_D &= 8.4 \text{ mA} * \left(1 - \frac{-1.5 \text{ V}}{-3 \text{ V}}\right)^2 \\ &= 8.4 \text{ mA} * (1 - 0.5)^2 \\ &= 4.2 \text{ mA} \end{aligned}$$

The transconductance g_m is defined as the ratio of the change in drain current to the change in gate voltage, at a constant drain-source voltage. In this case, the change in gate voltage is 1.5 V, and the change in drain current is 4.2 mA - 8.4 mA = -4.2 mA. So, the transconductance g_m is:

$$g_m = \frac{-4.2 \text{ mA}}{-1.5 \text{ V}} = 2.8 \text{ mA/V}$$

Therefore, the values of I_D and g_m are 4.2 mA and 2.8 mA/V, respectively.

3.b) State any three differences between JFET and BJT.

Below is a tabular comparison of three key differences between JFET (Junction Field Effect Transistor) and BJT (Bipolar Junction Transistor):

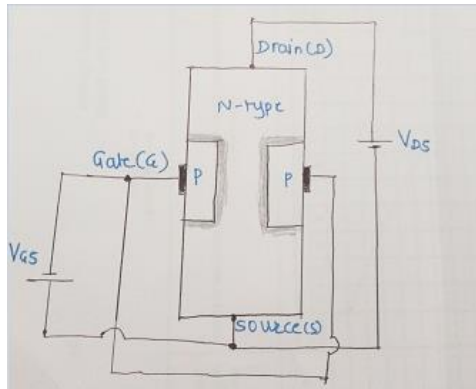
Feature	JFET (Junction Field Effect Transistor)	BJT (Bipolar Junction Transistor)
Type of Transistor	Unipolar Device: Utilizes only one type of charge carriers (either electrons or holes)	Bipolar Device: Uses both types of charge carriers (electrons and holes)
Control Mechanism	Voltage-controlled device: The current is controlled by the voltage applied to the gate terminal	Current-controlled device: The current is controlled by the current flowing into the base terminal
Voltage Polarity	JFET operates with reverse bias for normal operation. (N-channel JFET: Positive V_{gs} , P-channel JFET: Negative V_{gs})	BJT operates with forward bias for normal operation. (NPN: V_{be} is positive, PNP: V_{be} is negative)

These are some of the fundamental differences between JFET and BJT. Both types of transistors have their own applications and characteristics, making them suitable for different electronic circuits and purposes.

4.a) Explain the construction, principle of operation and Volt-Ampere characteristics of JFET.

Working of N-channel JFET

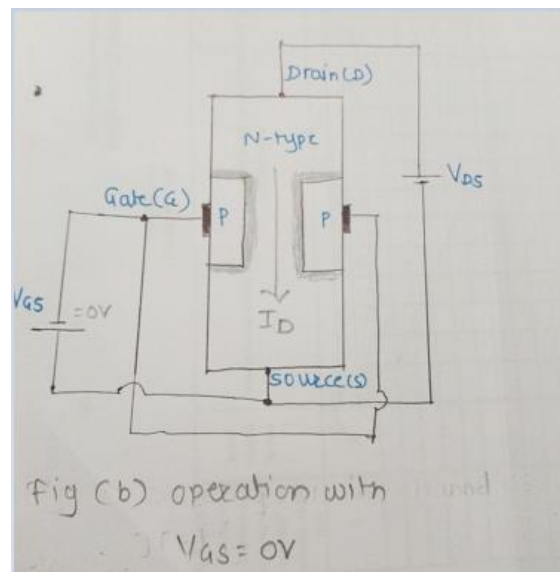
- Following fig. (a) shows biasing condition for working of N-channel JFET



- In above figure battery source voltage V_{GS} is applied between gate to source to make P-N junction reverse bias.
- A positive voltage is applied between drain and source terminal.
- We can consider following condition to observe working of N-channel JFET

A) when $V_{GS} = 0V$

- Following fig(b) shows the N-channel JFET when no bias is applied between gate and source

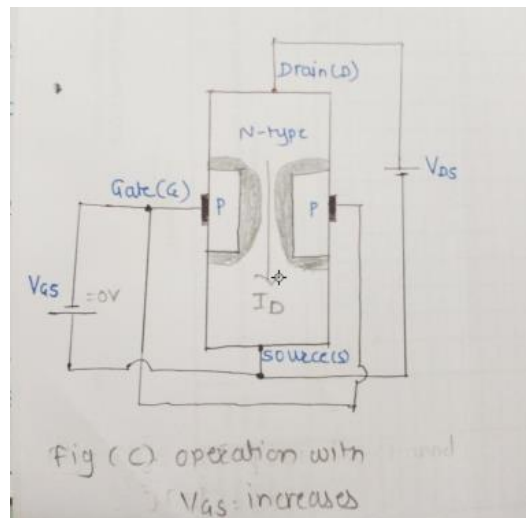


- When $V_{GS} = 0V$ and external voltage V_{DS} is applied between drain and source, the electrons flow from source to drain through the narrow channel.
- This constitutes the drain current (I_D) and its conventional direction is indicated from drain to source

B) When V_{GS} increased

- Following fig (c) shows the biasing of N channel JFET when V_{GS} has some finite value

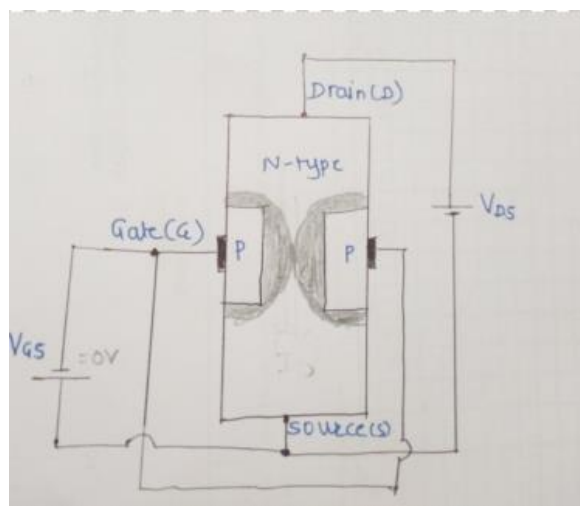
- V_{GS} is applied to make junction between gate to source reverse.



- When this reverse bias voltage V_{GS} is increased the depletion region becomes more wide. This reduces the effective width of the channel and therefore controls the flow of current through channel.
- When gate to source voltage increased further a stage is reached at which two depletion region touch each other as shown in fig (d).

At this stage the channel is completely blocked or pinch off and drain current reduced to zero

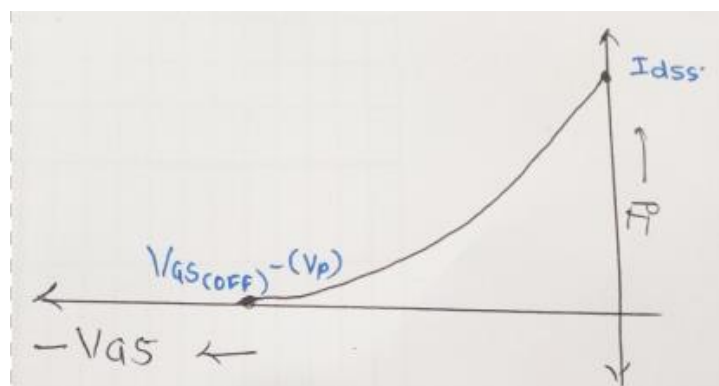
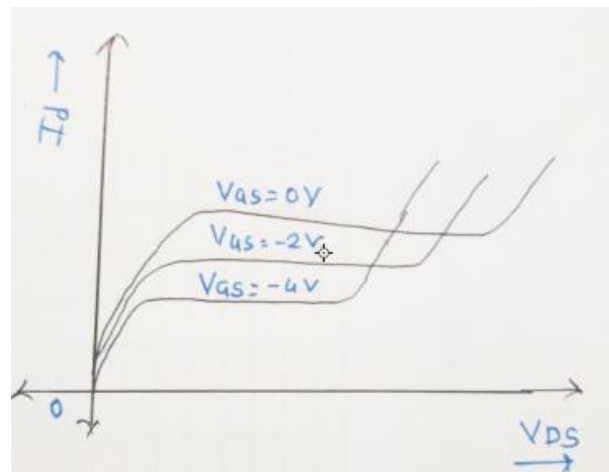
- The gate to source voltage at which the drain current is zero is called as pinch off voltage (V_P).



Symbol

- The following fig (e) shows schematic symbol for N-channel JFET
- In symbol vertical line represents N channel

- Characteristics of N channel JFET



4.b) A certain JFET operates in the linear region with a constant drain voltage of 1V. When the gate voltage is 2V, a drain current of 10mA flows, but when gate voltage is changed to 1V, the drain current becomes 22.8mA. Find (i) the pinch-off voltage (ii) the channel resistance for zero gate voltage.

Here are the solutions to the given problem:

(i) Pinch-off voltage

The pinch-off voltage is the gate voltage at which the drain current is zero. In this case, when the gate voltage is 2 V, the drain current is 10 mA. When the gate voltage is increased to 1 V, the drain current increases to 22.8 mA. This means that the pinch-off voltage is between 1 V and 2 V.

To find the pinch-off voltage, we can use the following equation:

$$I_D = I_{DSS} / (1 - V_{GS} / V_P)^2$$

where I_D is the drain current, I_{DSS} is the saturation current, V_{GS} is the gate-source voltage, and V_P is the pinch-off voltage.

Substituting the given values, we get:

$$10 \text{ mA} = I_{DSS} / (1 - 2 V / V_P)^2$$

$$10 \text{ mA} * (1 - 2 V / V_P)^2 = I_{DSS}$$

$$10 \text{ mA} * (1 - 4 / V_P)^2 = I_{DSS}$$

$$(10 \text{ mA} * V_P^2) / 16 = I_{DSS}$$

$$V_P^2 = 16 * I_{DSS} / 10 \text{ mA}$$

$$V_P^2 = 16 * 8.4 \text{ mA} / 10 \text{ mA}$$

$$V_P^2 = 134.4$$

$$V_P = 11.6 \text{ V}$$

Therefore, the pinch-off voltage is 11.6 V.

(ii) Channel resistance for zero gate voltage

The channel resistance for zero gate voltage is the resistance of the channel when the gate voltage is zero. In this case, the gate voltage is 0 V, and the drain current is 22.8 mA.

The channel resistance can be calculated using the following equation:

$$R_{DS} = V_{DS} / I_D$$

where R_{DS} is the channel resistance, V_{DS} is the drain-source voltage, and I_D is the drain current.

Substituting the given values, we get:

$$R_{DS} = 1 \text{ V} / 22.8 \text{ mA}$$

$$R_{DS} = 43.8 \text{ ohms}$$

Therefore, the channel resistance for zero gate voltage is 43.8 ohms.

5.a) Differentiate between a BJT and FET.

Below is a tabular comparison of the main differences between BJT (Bipolar Junction Transistor) and FET (Field Effect Transistor):

Feature	BJT (Bipolar Junction Transistor)	FET (Field Effect Transistor)
Charge Carriers	Utilizes both types of charge carriers (electrons and holes)	Utilizes only one type of charge carriers (either electrons or holes)
Control Mechanism	Current-controlled device: The current is controlled by base current (I_b)	Voltage-controlled device: The current is controlled by gate-source voltage (V_{gs})
Structure	Comprises of two PN junctions (Base-Emitter and Base-Collector)	Comprises of a gate terminal and a semiconductor channel
Voltage Polarity	BJT operates with forward bias for normal operation. (NPN: V_{be} is positive, PNP: V_{be} is negative)	FET operates with reverse bias for normal operation. (N-channel FET: Positive V_{gs} , P-channel FET: Negative V_{gs})
Input Impedance	Lower input impedance compared to FET	Higher input impedance compared to BJT

Feature	BJT (Bipolar Junction Transistor)	FET (Field Effect Transistor)
Output Impedance	Higher output impedance compared to FET	Lower output impedance compared to BJT
Applications	Commonly used in low to medium frequency applications	Commonly used in high-frequency and high-input impedance applications
Operating Principle	In BJT, both majority and minority charge carriers contribute to current flow	In FET, only majority charge carriers (either electrons or holes) contribute to current flow
Current Gain (β or H_{fe})	BJT typically has higher current gain (β) values	FET usually has lower current gain compared to BJT

These are some of the fundamental differences between BJT and FET. Both types of transistors have their own advantages and are used in various electronic circuits based on their specific characteristics and applications.

5.b) Explain the operation of FET with its characteristics and explain the different regions in transfer characteristics.

Here is an explanation of the operation of FET with its characteristics and the different regions in transfer characteristics:

A FET (Field Effect Transistor) is a type of transistor that is controlled by an electric field. It is a voltage-controlled device, which means that the

current flow between the drain and source terminals is controlled by the voltage applied to the gate terminal. The gate terminal is insulated from the channel, so no current flows through it. The voltage applied to the gate creates an electric field that affects the flow of charge carriers in the channel.

There are two main types of FETs: JFETs (junction field effect transistors) and MOSFETs (metal-oxide semiconductor field effect transistors). JFETs are made of semiconductor material, and MOSFETs are made of metal-oxide semiconductor material.

The operation of a FET can be divided into three regions:

- Cut-off region: This is the region where the gate voltage is so high that the depletion region completely blocks the flow of current through the channel.
- Ohmic region: This is the region where the gate voltage is low enough to allow some current to flow through the channel. The current through the channel is proportional to the gate voltage.
- Saturation region: This is the region where the gate voltage is so high that the depletion region is so wide that the channel is almost completely pinched off. The current through the channel is almost constant, regardless of the gate voltage.

The transfer characteristics of a FET are the curves that show the relationship between the drain current, I_D , and the gate-source voltage, V_{GS} . The transfer characteristics of a FET can be divided into three regions, corresponding to the three regions of operation:

- Cut-off region: The transfer characteristic is a horizontal line at zero current.
- Ohmic region: The transfer characteristic is a curve that increases linearly with V_{GS} .
- Saturation region: The transfer characteristic is a curve that levels off at a constant value of I_D .

The transfer characteristics of a FET are important for understanding how the transistor works and for designing circuits with FETs.

Here are some of the characteristics of FETs:

- They have a high input impedance, which means that they do not draw much current from the source.
- They have a low noise level, which makes them well-suited for applications where noise is a concern.
- They can be operated at high frequencies, which makes them suitable for applications such as radio and telecommunications.

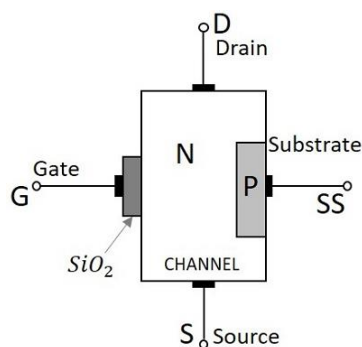
FETs are a versatile and powerful type of transistor that can be used in a wide variety of applications. Their voltage-controlled nature makes them particularly well-suited for applications where high input impedance, low noise, and high frequency operation are required.

6) Explain the construction of MOSFET and write down principle of operation of JFET?

Here is an explanation of the construction of MOSFET and the principle of operation of JFET:

Construction of MOSFET A MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is a type of transistor that is made of semiconductor material and metal-oxide semiconductor material. It is a voltage-controlled device, which means that the current flow between the drain and source terminals is controlled by the voltage applied to the gate terminal. The gate terminal is insulated from the channel, so no current flows through it. The voltage applied to the gate creates an electric field that affects the flow of charge carriers in the channel.

The construction of a MOSFET is shown below:



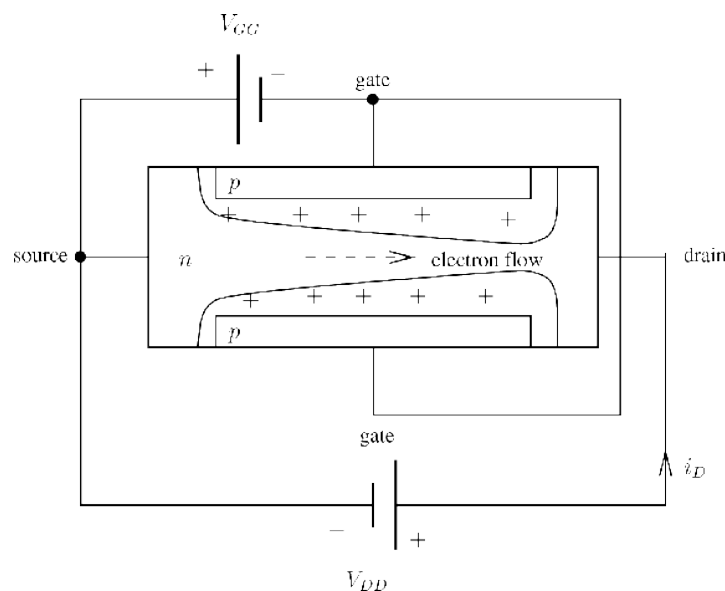
The MOSFET has three terminals: source, drain, and gate. The source and drain terminals are connected to the semiconductor material, and the gate terminal is connected to the metal-oxide semiconductor material.

The channel is a region of semiconductor material that is sandwiched between the source and drain terminals. The channel is either N-type or P-type, depending on the type of MOSFET.

The oxide layer is a thin layer of insulator that separates the gate terminal from the channel. The oxide layer prevents current from flowing between the gate terminal and the channel.

Principle of Operation of JFET

The principle of operation of a JFET is based on the electric field that is created by the gate voltage. When a positive voltage is applied to the gate, it creates an electric field that attracts electrons to the gate. This creates a depletion region in the channel, which is a region that is depleted of charge carriers.

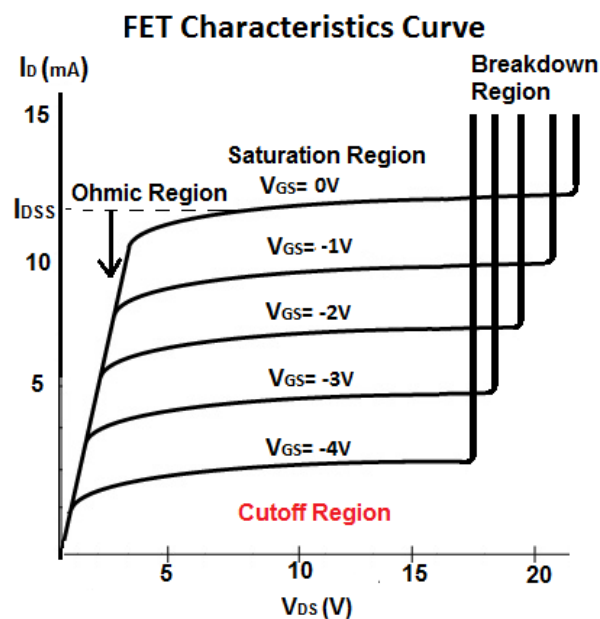


The depletion region acts as a barrier to the flow of current through the channel. The larger the gate voltage, the wider the depletion region, and the smaller the current that can flow through the channel.

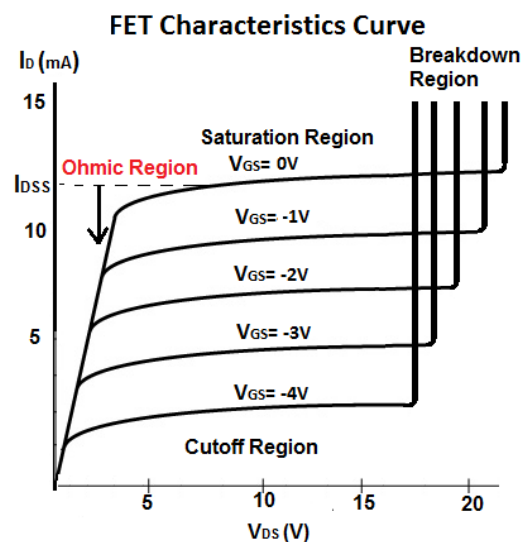
The drain current is controlled by the gate voltage. When the gate voltage is zero, the depletion region is very narrow, and the maximum current can flow through the channel. As the gate voltage is increased, the depletion region widens, and the current through the channel decreases.

The JFET can be operated in three different regions:

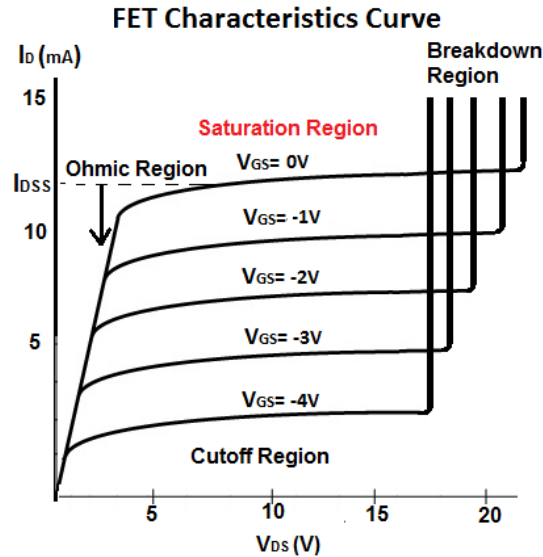
- Cut-off region: This is the region where the gate voltage is so high that the depletion region completely blocks the flow of current through the channel.



- Ohmic region: This is the region where the gate voltage is low enough to allow some current to flow through the channel. The current through the channel is proportional to the gate voltage.



- Saturation region: This is the region where the gate voltage is so high that the depletion region is so wide that the channel is almost completely pinched off. The current through the channel is almost constant, regardless of the gate voltage.



The JFET is a versatile and powerful type of transistor that can be used in a wide variety of applications. Its voltage-controlled nature makes it particularly well-suited for applications where high input impedance, low noise, and high frequency operation are required.

7) Explain MOSFET as a Capacitor

Here is an explanation of MOSFET as a capacitor with neat diagrams:

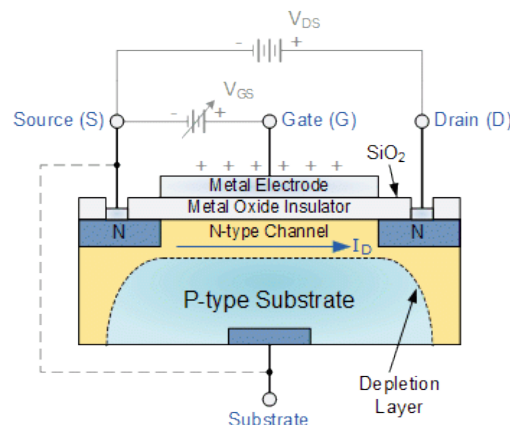
A MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) can be used as a capacitor because the gate oxide layer behaves like a capacitor. The gate oxide layer is a thin layer of insulator that separates the gate terminal from the channel. The oxide layer prevents current from flowing between the gate terminal and the channel, but it does allow an electric field to pass through.

The capacitance of a MOSFET is determined by the following factors:

- The area of the gate oxide layer
- The thickness of the gate oxide layer
- The dielectric constant of the gate oxide material

The capacitance of a MOSFET can be used in a variety of applications, such as:

- Charge storage
- Frequency filtering
- Voltage sensing



The gate oxide layer of a MOSFET is typically very thin, on the order of a few nanometers. This makes the capacitance of a MOSFET very high, typically in the range of femtofarads (fF).

The capacitance of a MOSFET can be increased by increasing the area of the gate oxide layer. This can be done by making the gate terminal larger or by using a MOSFET with a multi-gate structure.

The capacitance of a MOSFET can be decreased by decreasing the thickness of the gate oxide layer. However, the thickness of the gate oxide layer is limited by the need to prevent current from flowing between the gate terminal and the channel.

The capacitance of a MOSFET is a very important property that can be used in a variety of applications. By understanding the capacitance of a MOSFET, you can design circuits that take advantage of this property.

UNIT 5

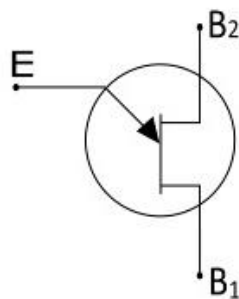
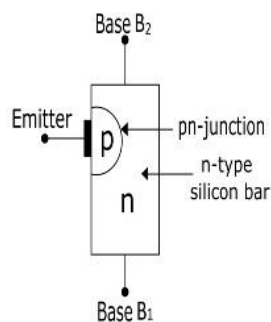
1.a) Describe the construction and working of UJT with its equivalent Circuit and V-I Characteristics.

A Unijunction Transistor (UJT) is a three-terminal semiconductor device. The main characteristics of UJT is when it is triggered, the emitter current increases re-generatively until it is limited by emitter power supply. Due to this characteristic feature, it is used in applications like switching pulse generator, saw-tooth wave generator etc.

Construction of UJT

The UJT consists of an n-type silicon semiconductor bar with an electrical on each end. The terminals of these connections are called Base terminals (B_1 and B_2). Near to base B_2 , a pn-junction is formed between a p-type emitter and the n-type silicon bar. The terminal of this junction is called emitter terminal (E).

Since the device has three terminals and one pn-junction, for this region this is called as a Unijunction Transistor (UJT).



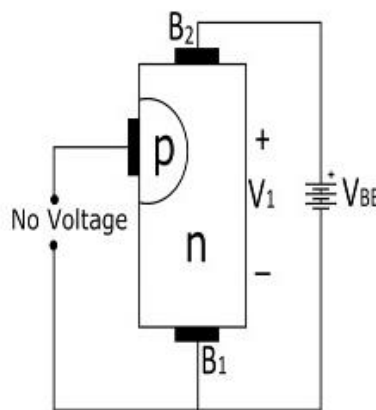
The device has only pn-junction so it forms a diode. Because the two base leads are taken from one section of the diode, hence the device is also called as Double-Based Diode.

The emitter is heavily doped while the n-region is lightly doped. Thus, the resistance between base terminals is very high when emitter terminal is open.

Operation of UJT

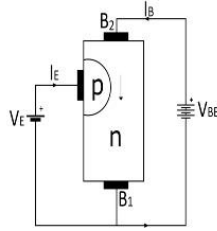
With Emitter Open

When the voltage V_{BB} is applied with emitter open. A potential gradient is established along the n-type silicon bar. As the emitter is located close to the base B_2 , thus a major part of V_{BB} appears between the emitter and base B_1 . The voltage V_1 between emitter and B_1 , establishes a reverse bias on the pn-junction and the emitter current is cut off, but a small leakage current flows from B_2 to emitter due to minority charge carriers. Thus, the device is said to be in OFF state.

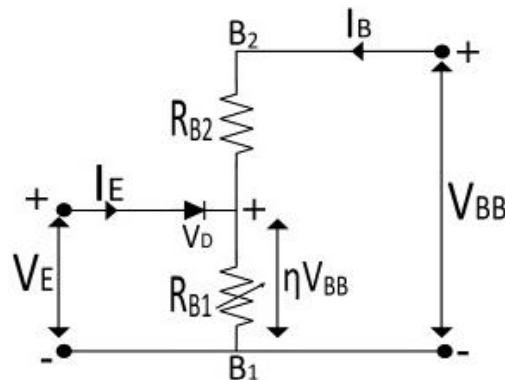


With Emitter at Positive Potential

When a positive voltage is applied at the emitter terminal, the pn-junction will remain reverse biased till the input voltage is less than V_1 . As soon as the input voltage at emitter exceeds V_1 , the pn-junction becomes forward biased. Under this condition, holes are supplied from p-type region into the n-type bar. These holes are repelled by positive B_2 terminal and attracted towards the B_1 terminal. This increase in the number of holes in the emitter to B_1 region results in the decrease of resistance of this section of the bar. Because of this, the internal voltage drop from emitter to B_1 region is reduced, thus the emitter current (I_E) increases. As more holes are supplied, a condition of saturation is reached. At the point of saturation, the emitter current is limited by the emitter power supply. Now, the device is conducting, hence said to be in ON state.



Equivalent Circuit of UJT



- The resistance of silicon bar is called as the inter-base resistance (has a value from 4 kΩ to 10 kΩ).
- The resistance R_{B1} is the resistance of the bar between emitter and B_1 region. The value of this is variable and depends upon the bias voltage across the pn-junction.
- The resistance R_{B2} is the resistance of the bar between emitter and B_2 region.
- The emitter pn-junction is represented by a diode.
- With no voltage applied to the UJT, the value of inter-base resistance is given by

$$R_{BB} = R_{B1} + R_{B2}$$

- The intrinsic stand-off ration (η) of UJT is given by

$$\eta = \frac{V_1}{V_{BB}} = \frac{R_{B1}}{R_{B1} + R_{B2}}$$

The voltage across R_{B1} is

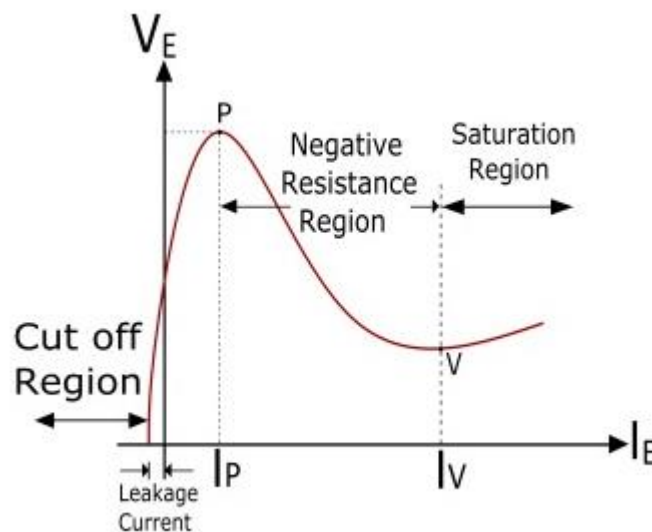
$$V_1 = \frac{R_{B1}}{R_{B1} + R_{B2}} V_{BB} = \eta V_{BB}$$

- The value of η generally lies between 0.51 and 0.82.
- The Peak Point Voltage (V_P) of the UJT

$$V_1 = \eta V_{BB} + V_D$$

Characteristics of UJT

The curve between emitter voltage (V_E) and emitter current (I_E) of UJT, at a given value of V_{BB} is known as emitter characteristics of UJT.



Important points from the characteristics are –

- At first, in the cut off region, when the emitter voltage increases from zero, due to the minority charge carriers, a small current flows from terminal B_2 to emitter. This is called as leakage current.
- Above the definite value of V_E , the emitter current (I_E) starts to flow and increases until the peak (V_P and I_P) is reached at point P.
- After point P, an increase in V_E causes a sudden increase in I_E with a corresponding decrease in V_E . This is the Negative Resistance Region of the curve as with the increase in I_E , V_E decreases.
- The negative resistance region of the curve ends at the valley-point (V), having valley-point voltage V_V and current I_V . After the valley-point the device is driven to saturation.

Advantages of UJT

- Low cost
- Excellent characteristics
- Low power absorbing device under normal operating conditions

Applications of UJT

- Oscillators
- Trigger Circuits
- Saw tooth generator
- Bi-stable networks

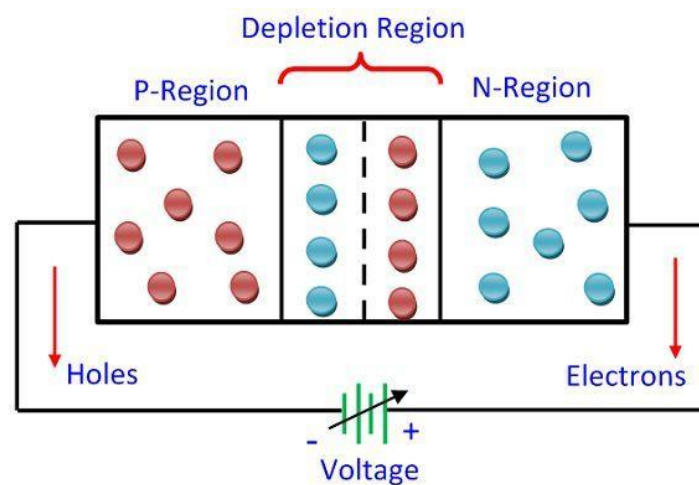
- Pulse and voltage sensing circuits
- UJT relaxation oscillators
- Over voltage detectors

1.b) With a neat diagram explain the working of Varactor Diode

Here is an explanation of the working of a varactor diode with a neat diagram:

A varactor diode is a type of diode that can be used as a variable capacitor. The capacitance of a varactor diode is inversely proportional to the square root of the applied voltage. This means that as the applied voltage increases, the capacitance decreases.

The working of a varactor diode is based on the depletion region that is created by the reverse bias voltage. The depletion region is a region that is depleted of charge carriers. The width of the depletion region is determined by the applied voltage.



Depletion Region in a Reverse Biased P-N junction

When a reverse bias voltage is applied to a varactor diode, the depletion region widens. This decreases the capacitance of the diode. When the reverse bias voltage is decreased, the depletion region narrows. This increases the capacitance of the diode.

The capacitance of a varactor diode can be used in a variety of applications, such as:

- Frequency tuning
- Voltage-controlled oscillators
- Phase-locked loops

Varactor diodes are a versatile and powerful type of diode that can be used in a variety of applications. By understanding the working of a varactor diode, you can design circuits that take advantage of this property.

Here are some additional details about varactor diodes:

- Varactor diodes are also known as varicaps or tuning diodes.
- The capacitance of a varactor diode is typically in the range of picofarads (pF).
- Varactor diodes are often used in conjunction with resonant circuits to tune the frequency of the circuit.
- Varactor diodes are also used in voltage-controlled oscillators (VCOs) and phase-locked loops (PLLs).

2.a) Explain the working principle of Tunnel diode and explain how it works as an Oscillator?

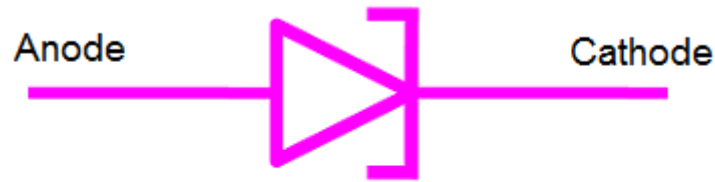
Tunnel diode definition

A Tunnel diode is a heavily doped p-n junction diode in which the electric current decreases as the voltage increases.

In tunnel diode, electric current is caused by “Tunneling”. The tunnel diode is used as a very fast switching device in computers. It is also used in high-frequency oscillators and amplifiers.

Symbol of tunnel diode

The circuit symbol of tunnel diode is shown in the below figure. In tunnel diode, the p-type semiconductor act as an anode and the n-type semiconductor act as a cathode.



Tunnel diode symbol

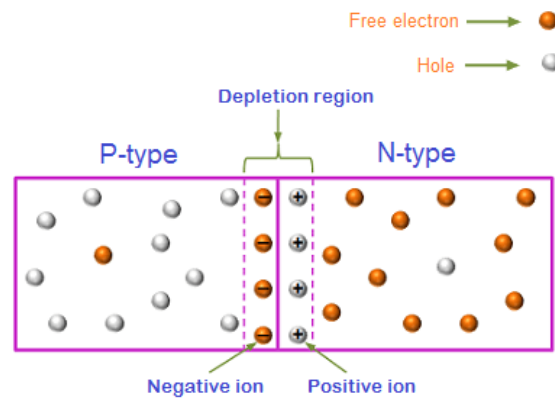
We know that a anode is a positively charged electrode which attracts electrons whereas cathode is a negatively charged electrode which emits electrons. In tunnel diode, n-type semiconductor emits or produces electrons so it is referred to as the cathode. On the other hand, p-type semiconductor attracts electrons emitted from the n-type semiconductor so p-type semiconductor is referred to as the anode.

What is a tunnel diode

Tunnel diodes are one of the most significant solid-state electronic devices which have made their appearance in the last decade. Tunnel diode was invented in 1958 by Leo Esaki.

Leo Esaki observed that if a semiconductor diode is heavily doped with impurities, it will exhibit negative resistance. Negative resistance means the current across the tunnel diode decreases when the voltage increases. In 1973 Leo Esaki received the Nobel Prize in physics for discovering the electron tunneling effect used in these diodes.

A tunnel diode is also known as Esaki diode which is named after Leo Esaki for his work on the tunneling effect. The operation of tunnel diode depends on the quantum mechanics principle known as “Tunneling”. In electronics, tunneling means a direct flow of electrons across the small depletion region from n-side conduction band into the p-side valence band.



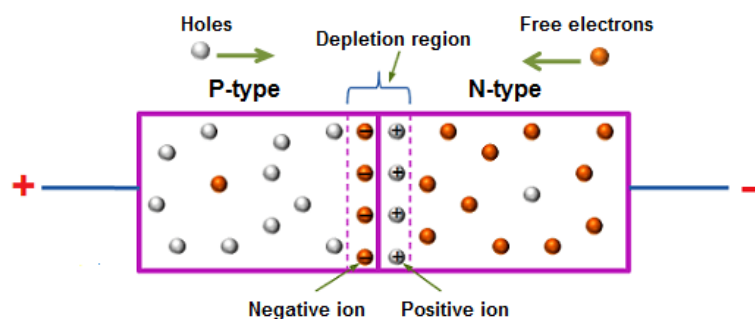
Tunnel diode

The germanium material is commonly used to make the tunnel diodes. They are also made from other types of materials such as gallium arsenide, gallium antimonide, and silicon.

Width of the depletion region in tunnel diode

The depletion region is a region in a p-n junction diode where mobile charge carriers (free electrons and holes) are absent. Depletion region acts like a barrier that opposes the flow of electrons from the n-type semiconductor and holes from the p-type semiconductor.

The width of a depletion region depends on the number of impurities added. Impurities are the atoms introduced into the p-type and n-type semiconductor to increase electrical conductivity. If a small number of impurities are added to the p-n junction diode (p-type and n-type semiconductor), a wide depletion region is formed. On the other hand, if large number of impurities are added to the p-n junction diode, a narrow depletion region is formed.



Forward bias tunnel diode

In tunnel diode, the p-type and n-type semiconductor is heavily doped which means a large number of impurities are introduced into the p-type and n-type semiconductor. This heavy doping process produces an extremely narrow depletion region. The concentration of impurities in tunnel diode is 1000 times greater than the normal p-n junction diode.

In normal p-n junction diode, the depletion width is large as compared to the tunnel diode. This wide depletion layer or depletion region in normal diode opposes the flow of current. Hence, depletion layer acts as a barrier. To overcome this barrier, we need to apply sufficient voltage. When sufficient voltage is applied, electric current starts flowing through the normal p-n junction diode.

Unlike the normal p-n junction diode, the width of a depletion layer in tunnel diode is extremely narrow. So applying a small voltage is enough to produce electric current in tunnel diode.

Tunnel diodes are capable of remaining stable for a long duration of time than the ordinary p-n junction diodes. They are also capable of high-speed operations.

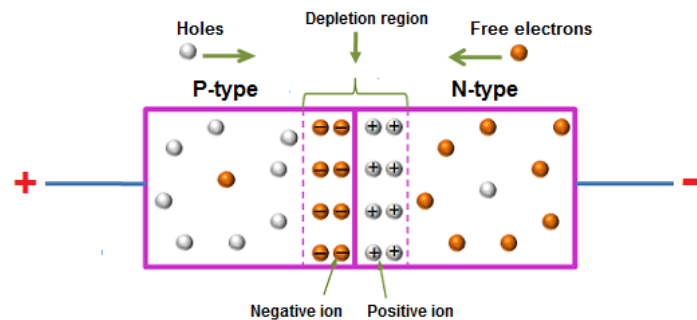
The depletion region or depletion layer in a p-n junction diode is made up of positive ions and negative ions. Because of these positive and negative ions, there exists a built-in-potential or electric field in the depletion region. This electric field in the depletion region exerts electric force in a direction opposite to that of the external electric field (voltage).

Another thing we need to remember is that the valence band and conduction band energy levels in the n-type semiconductor are slightly lower than the valence band and conduction band energy levels in the p-type semiconductor. This difference in energy levels is due to the differences in the energy levels of the dopant atoms (donor or acceptor atoms) used to form the n-type and p-type semiconductor.

Electric current in ordinary p-n junction diode

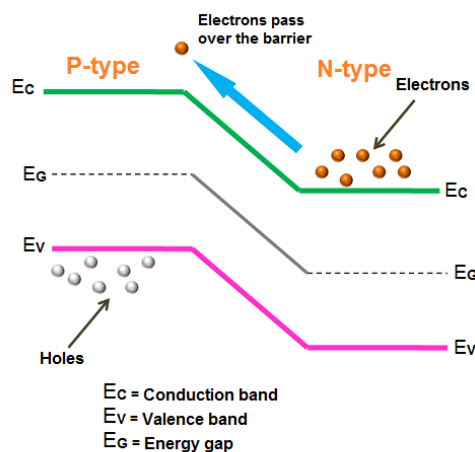
When a forward bias voltage is applied to the ordinary p-n junction diode, the width of depletion region decreases and at the same time the barrier height also decreases. However, the electrons in the n-type semiconductor

cannot penetrate through the depletion layer because the built-in voltage of depletion layer opposes the flow of electrons.

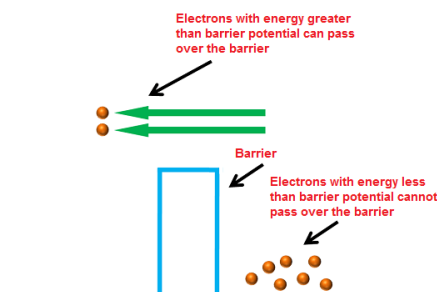


Forward bias p-n junction diode

If the applied voltage is greater than the built-in voltage of depletion layer, the electrons from n-side overcomes the opposing force from depletion layer and then enters into p-side. In simple words, the electrons can pass over the barrier (depletion layer) if the energy of the electrons is greater than the barrier height or barrier potential.



Ordinary P-N junction diode

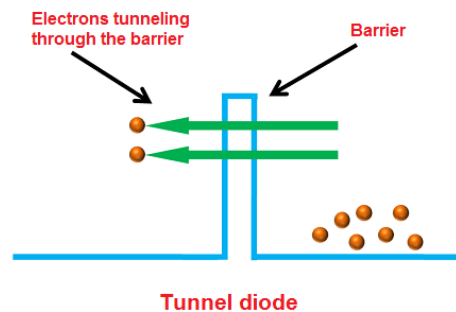


Ordinary p-n junction diode

Therefore, an ordinary p-n junction diode produces electric current only if the applied voltage is greater than the built-in voltage of the depletion region.

Electric current in tunnel diode

In tunnel diode, the valence band and conduction band energy levels in the n-type semiconductor are lower than the valence band and conduction band energy levels in the p-type semiconductor. Unlike the ordinary p-n junction diode, the difference in energy levels is very high in tunnel diode. Because of this high difference in energy levels, the conduction band of the n-type material overlaps with the valence band of the p-type material.

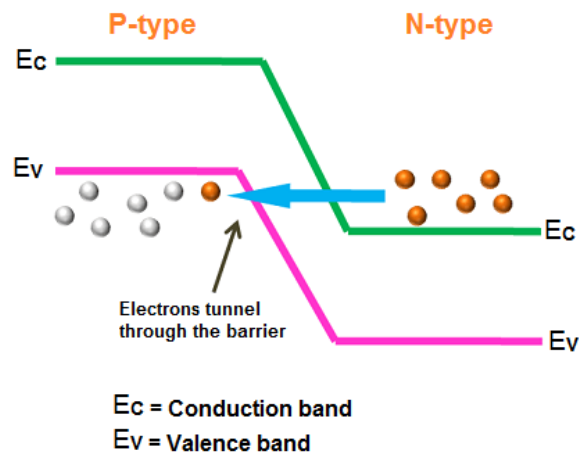


Quantum mechanics says that the electrons will directly penetrate through the depletion layer or barrier if the depletion width is very small.

The depletion layer of tunnel diode is very small. It is in nanometers. So the electrons can directly tunnel across the small depletion region from n-side conduction band into the p-side valence band.

In ordinary diodes, current is produced when the applied voltage is greater than the built-in voltage of the depletion region. But in tunnel diodes, a small voltage which is less than the built-in voltage of depletion region is enough to produce electric current.

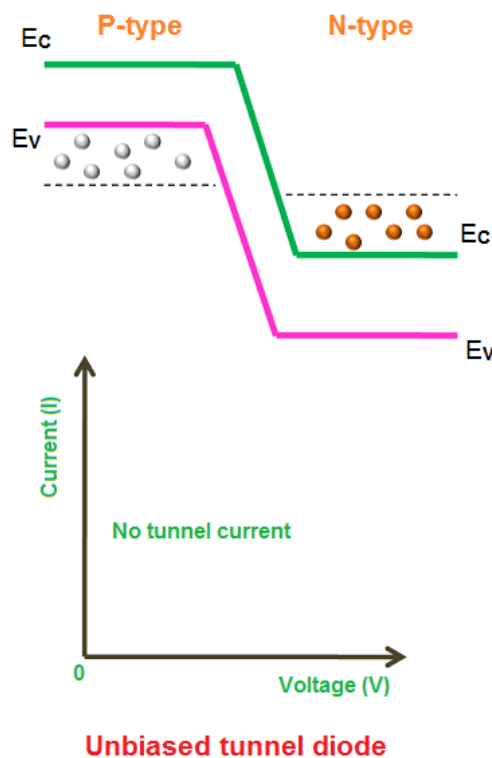
In tunnel diodes, the electrons need not overcome the opposing force from the depletion layer to produce electric current. The electrons can directly tunnel from the conduction band of n-region into the valence band of p-region. Thus, electric current is produced in tunnel diode.



How tunnel diode works?

Step 1: Unbiased tunnel diode

When no voltage is applied to the tunnel diode, it is said to be an unbiased tunnel diode. In tunnel diode, the conduction band of the n-type material overlaps with the valence band of the p-type material because of the heavy doping.



Because of this overlapping, the conduction band electrons at n-side and valence band holes at p-side are nearly at the same energy level. So when the temperature increases, some electrons tunnel from the conduction

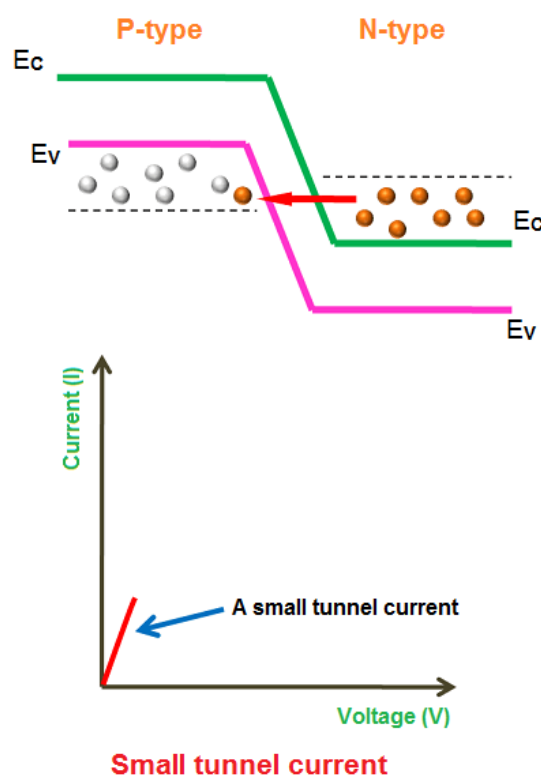
band of n-region to the valence band of p-region. In a similar way, holes tunnel from the valence band of p-region to the conduction band of n-region.

However, the net current flow will be zero because an equal number of charge carriers (free electrons and holes) flow in opposite directions.

Step 2: Small voltage applied to the tunnel diode

When a small voltage is applied to the tunnel diode which is less than the built-in voltage of the depletion layer, no forward current flows through the junction.

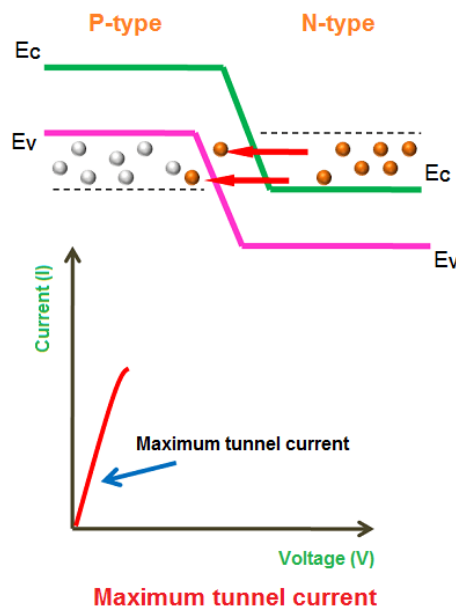
However, a small number of electrons in the conduction band of the n-region will tunnel to the empty states of the valence band in p-region. This will create a small forward bias tunnel current. Thus, tunnel current starts flowing with a small application of voltage.



Step 3: Applied voltage is slightly increased

When the voltage applied to the tunnel diode is slightly increased, a large number of free electrons at n-side and holes at p-side are generated.

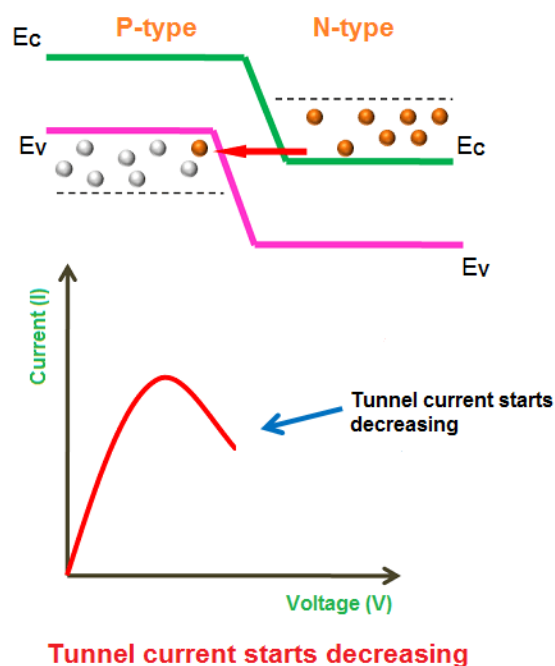
Because of the increase in voltage, the overlapping of the conduction band and valence band is increased.



In simple words, the energy level of an n-side conduction band becomes exactly equal to the energy level of a p-side valence band. As a result, maximum tunnel current flows.

Step 4: Applied voltage is further increased

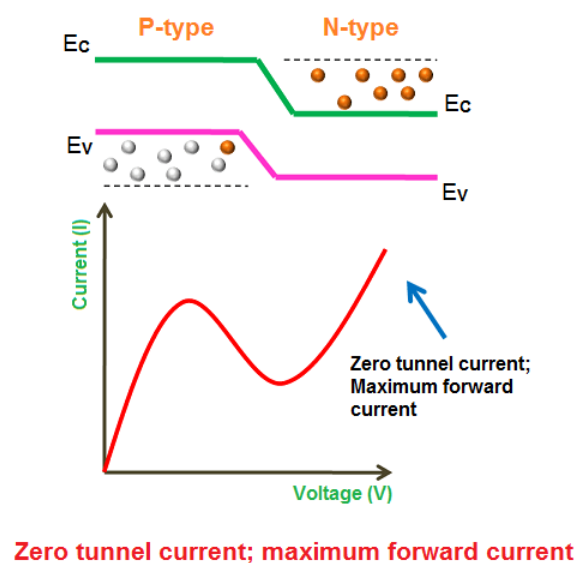
If the applied voltage is further increased, a slight misalign of the conduction band and valence band takes place.



Since the conduction band of the n-type material and the valence band of the p-type material still overlap. The electrons tunnel from the conduction band of n-region to the valence band of p-region and cause a small current flow. Thus, the tunneling current starts decreasing.

Step 5: Applied voltage is largely increased

If the applied voltage is largely increased, the tunneling current drops to zero. At this point, the conduction band and valence band no longer overlap and the tunnel diode operates in the same manner as a normal p-n junction diode.



If this applied voltage is greater than the built-in potential of the depletion layer, the regular forward current starts flowing through the tunnel diode.

The portion of the curve in which current decreases as the voltage increases is the negative resistance region of the tunnel diode. The negative resistance region is the most important and most widely used characteristic of the tunnel diode.

A tunnel diode operating in the negative resistance region can be used as an amplifier or an oscillator.

Advantages of tunnel diodes

- Long life
- High-speed operation

- Low noise
- Low power consumption

Disadvantages of tunnel diodes

- Tunnel diodes cannot be fabricated in large numbers
- Being a two terminal device, the input and output are not isolated from one another.

Applications of tunnel diodes

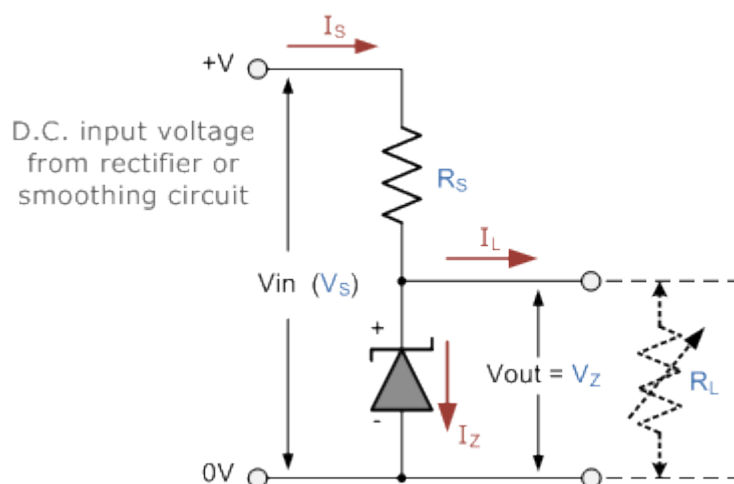
- Tunnel diodes are used as logic memory storage devices.
- Tunnel diodes are used in relaxation oscillator circuits.
- Tunnel diode is used as an ultra high-speed switch.
- Tunnel diodes are used in FM receivers.

2.b) Explain how the zener diode is used for regulation purpose.

Here is an explanation of how a Zener diode is used for regulation purpose:

A Zener diode is a type of diode that has a reverse breakdown voltage. This means that when the diode is reverse biased, it will conduct current at a specific voltage. This voltage is called the Zener voltage.

Zener diodes can be used to regulate voltage in a circuit. A simple Zener diode voltage regulator circuit is shown below:



In this circuit, the Zener diode is connected in parallel with the load. The resistor is used to limit the current through the Zener diode.

When the input voltage is below the Zener voltage, the Zener diode is reverse biased and does not conduct current. The entire input voltage is applied to the load.

When the input voltage is above the Zener voltage, the Zener diode conducts current and the voltage across the Zener diode is equal to the Zener voltage. The voltage across the load is also equal to the Zener voltage.

This means that the Zener diode regulates the voltage across the load to the Zener voltage.

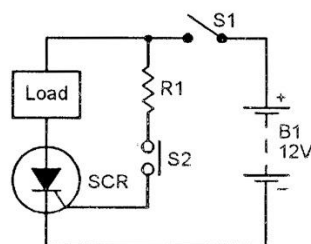
Zener diodes are a versatile and powerful type of diode that can be used in a variety of applications. By understanding how a Zener diode works, you can design circuits that take advantage of this property.

Here are some additional details about Zener diodes:

- Zener diodes are also known as breakdown diodes or voltage regulators.
- The Zener voltage of a Zener diode is typically in the range of 2 to 200 volts.
- Zener diodes are often used in conjunction with resistors to create voltage regulators.
- Zener diodes are also used in a variety of other applications, such as frequency doubling and wave shaping.

3.a) Draw the circuit diagram of SCR and explain its operation along with its characteristics.

Here is a diagram of a SCR (Silicon Controlled Rectifier) and an explanation of its operation along with its characteristics:



The SCR is a three-terminal semiconductor device that can be used as a switch. The terminals are called anode, gate, and cathode.

The SCR has two regions of operation:

- Cut-off region: This is the region where the SCR is off. No current flows through the SCR in this region.
- On region: This is the region where the SCR is on. Current flows through the SCR in this region.

The SCR can be turned on by applying a positive voltage to the gate terminal. This voltage must be greater than the gate threshold voltage, which is typically in the range of 0.5 to 1 volt.

Once the SCR is turned on, it will remain on even if the gate voltage is removed. The SCR can only be turned off by reducing the current below a certain value, called the holding current.

The SCR has the following characteristics:

- High input impedance: The input impedance of the SCR is very high, which means that it draws very little current from the gate terminal.
- Low output impedance: The output impedance of the SCR is very low, which means that it can sink a large amount of current.
- Bidirectional: The SCR can be used to conduct current in both directions.

The SCR is a versatile and powerful device that can be used in a variety of applications. It is often used in power control applications, such as motor control and light dimming.

Here are some additional details about SCRs:

- SCRs are also known as thyristors.
- SCRs are available in a variety of sizes and ratings.
- SCRs can be used to control high voltages and currents.
- SCRs are a reliable and robust device.

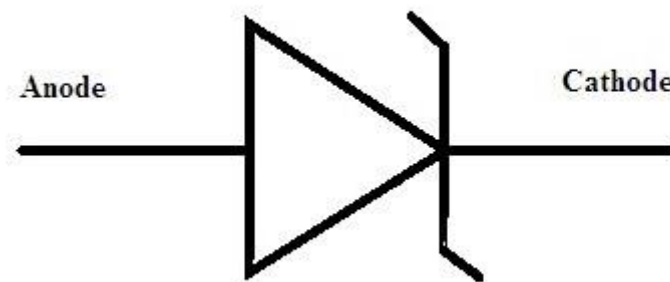
3.b) With a neat diagram explain the working and characteristics of Zener Diode.

Zener Diode Working and Its Characteristics

A basic semiconductor diode can operate in forward bias but it cannot conduct during reverse bias. The special type of diode that can conduct in reverse bias is referred to as a Zener diode. This diode can conduct in forward bias just like a normal diode. The junction for the Zener diode will be heavily doped. Because of this reason, it can work as normally as a basic junction in forwarding bias and can tolerate the reverse currents during reverse bias. The main purpose of it is to use in stabilizers.

What is a Zener Diode?

A diode in which the flow of current is from anode to cathode and cathode to anode defining that it has the capacity of conducting in both forward and reverse biases are referred to a Zener diode.



Symbol of Zener Diode

Working Principle:

- Usually, the condition of the diode with basic p-n as its junction during reverse bias is that there is no chance of conduction because the depletion region width is comparatively high.
- As the applied reverse voltage tends to increase that result in the increment of the width of the depletion region. Even there exist some minority carriers which gain some energy because of increment of reverse voltage.
- Due to the gain in kinetic energy of the minority carriers, these free electrons in movement collide with the stationary ions. This results in the formation of more free electrons.
- Further, these again collide with remaining stationary ions and this process continues it is referred to as carrier multiplication.

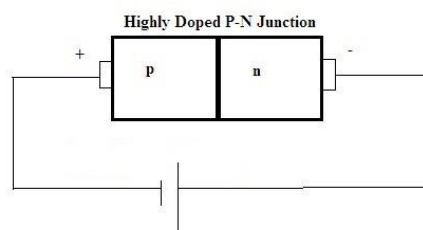
- Because of carrier multiplication, a huge multiple of free electrons are created and the complete region of the diode becomes conductive resulting in the breakdown known as avalanche breakdown.
- Generally, this is not the case in the Zener diode. In Zener diode, the junction is doped with the highest concentration. Because of this reason when the reverse voltage has applied the width of the depletion region tends to minimize.
- As there exist the maximum concentration of the impure atoms in it. It creates the maximum number of ions in it.
- As soon as the diode exceeds the threshold value the electrons that are in the covalent bonds tend to come out in the depletion region so that it can make depletion region conductive.
- Hence this type of breakdown is referred to as Zener breakdown. The occurrence of this breakdown will be at certain voltage termed as Zener voltage.
- Just as cut in voltage in normal diode here it is Zener voltage. Once the applied voltage exceeds the value of voltage it tends to conduct.
- This value of the Zener voltage is properly adjusted at the time of manufacturing by the proper concentration in doping.
- In case of occurring of break down further, there is no possibility of occurrence of avalanche breakdown.

This is the principle lying behind the functioning of the diode.

Please refer to this link to know more about Zener Diode MCQs

Zener Diode Circuit

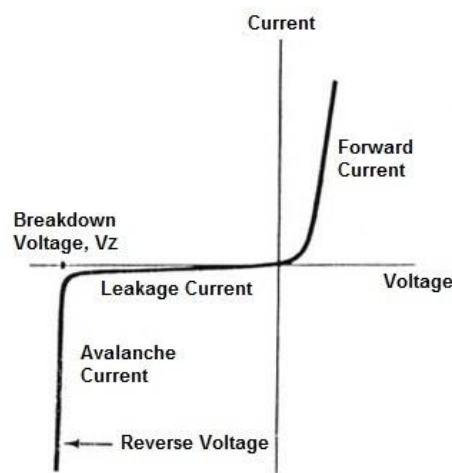
The circuit for this type of diode is very simple. One has to connect the highly doped junction in reverse bias. The circuit can be represented as follows.



Zener Diode Circuit

Characteristics

The graphical representation can give basic characteristics. These are normally termed as V-I characteristics.

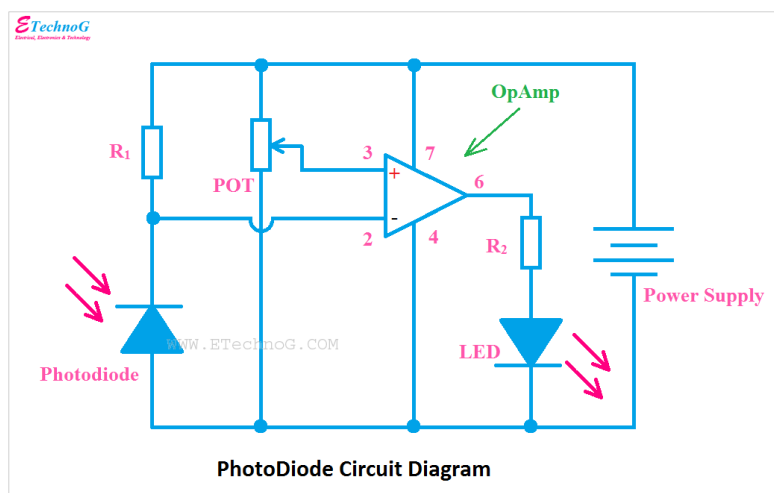


Zener Diode V-I Characteristics Curve

As per the above analysis, it is evident that during forwarding bias Zener diode characteristics will remain the same as that of the normal diode. Whereas after the applied voltage crosses the value of Zener voltage (V_Z) the Zener breakdown takes place. After the breakdown, the flow of the current in the circuit tends to increase immediately.

4.a) Draw the circuit diagram of photo diode and explain its operation along with its characteristics.

Here is a diagram of a photodiode and an explanation of its operation along with its characteristics:



The photodiode is a semiconductor device that converts light into electricity. It is made of a semiconductor material, such as silicon or germanium, with a p-n junction.

When light shines on the photodiode, it creates electron-hole pairs in the depletion region of the p-n junction. These electron-hole pairs are separated by the electric field in the depletion region, and the electrons flow to the n-type region and the holes flow to the p-type region. This creates a current that flows through the photodiode.

The amount of current that flows through the photodiode is proportional to the intensity of the light that shines on it. This means that the photodiode can be used to measure the intensity of light.

The photodiode has the following characteristics:

- High sensitivity: The photodiode is very sensitive to light, and it can be used to measure even very low levels of light.
- Wide bandwidth: The photodiode has a wide bandwidth, and it can be used to measure light over a wide range of frequencies.
- Low noise: The photodiode is a low-noise device, and it can be used to measure light with high accuracy.

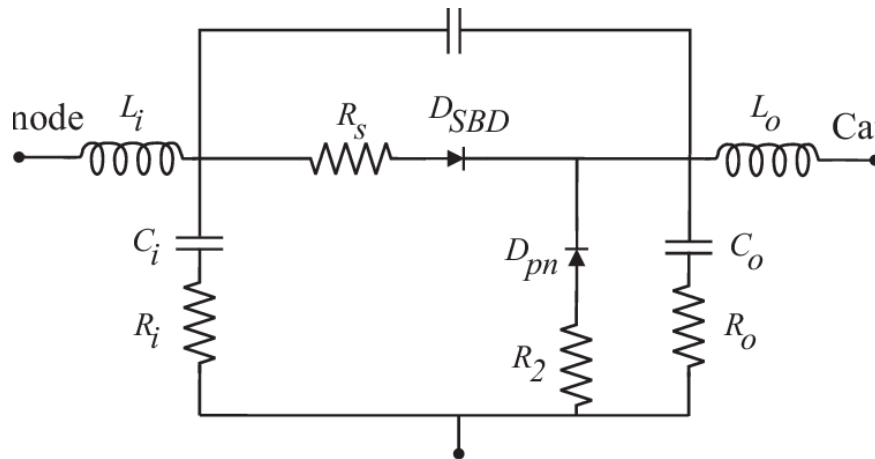
The photodiode is a versatile and powerful device that can be used in a variety of applications. It is often used in optical communication, light detection, and spectroscopy.

Here are some additional details about photodiodes:

- Photodiodes are also known as light-sensitive diodes or photoconductors.
- Photodiodes are available in a variety of sizes and sensitivities.
- Photodiodes can be used to measure light in a variety of wavelengths.
- Photodiodes are a reliable and robust device.

4.b) Draw the circuit diagram of Schottky diode and explain its operation along with its characteristics.

Here is a diagram of a Schottky diode and an explanation of its operation along with its characteristics:



The Schottky diode is a type of diode that is made of a metal-semiconductor junction. It is a fast-switching diode with a low forward voltage drop.

The Schottky diode is similar to a PN junction diode, but it has a different structure. In a PN junction diode, the junction is formed between a p-type semiconductor and an n-type semiconductor. In a Schottky diode, the junction is formed between a metal and a semiconductor.

The Schottky diode has the following characteristics:

- Low forward voltage drop: The forward voltage drop of a Schottky diode is typically in the range of 0.2 to 0.4 volts. This is much lower than the forward voltage drop of a PN junction diode.
- Fast switching: The Schottky diode is a fast-switching diode, which means that it can switch between the on and off states very quickly.
- Low noise: The Schottky diode is a low-noise diode, which means that it can be used in applications where noise is a concern.

The Schottky diode is a versatile and powerful device that can be used in a variety of applications. It is often used in digital circuits, RF circuits, and power electronics applications.

Here are some additional details about Schottky diodes:

- Schottky diodes are also known as hot carrier diodes or metal-semiconductor diodes.
- Schottky diodes are available in a variety of sizes and ratings.
- Schottky diodes can be used to switch high voltages and currents.
- Schottky diodes are a reliable and robust device.

5) Explain about LED.

LED

LED (Light Emitting Diode) is an optoelectronic device which works on the principle of electro-luminance. Electro-luminance is the property of the material to convert electrical energy into light energy and later it radiates this light energy. In the same way, the semiconductor in LED emits light under the influence of electric field.

The symbol of LED is formed by merging the symbol of P-N Junction diode and outward arrows. These outward arrows symbolise the light radiated by the light emitting diode.

Now, the question arises how the semiconductor material in LED emits light? The answer to this question lies in the construction and working of LED. The symbol of LED is described in the diagram below, the same symbol is used in electronics circuits.

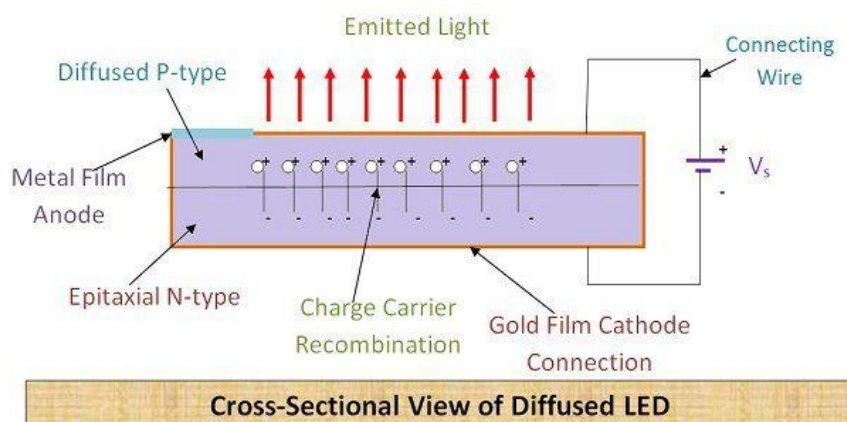
Construction of LED

The semiconductor material used in LED is Gallium Arsenide (GaAs), Gallium Phosphide (GaP) or Gallium Arsenide Phosphide (GaAsP). Any of the above-mentioned compounds can be used for the construction of LED, but the colour of radiated light changes with the change in material. Below are some of the material and their respective colour of light which they emit. In addition to it, the ranges of typical forward voltage are also given below.

MATERIALS IN CONSTRUCTION	COLOUR	FORWARD VOLTAGE (IN VOLTS)
GaP	Green/Red	2.2
GaAsP	Yellow	2.2
GaAsP	Red	1.8
GaN	White	4.1
GaN	Blue	5.0
AlInGaP	Amber	2.1
AlInGaP	Yellow	2.1

Internal Architecture of LED

The semiconductor layer of P-type is placed above N-type because the charge carrier recombination occurs in p-type. Besides, it is the surface of the device, thus, the light emitted can be easily seen on the surface. If P-type will be placed below the light will be emitted from the surface of P-type but we will not be able to see it. This is the reason that P-type is placed above.



The P-type layer is formed from diffusion of semiconductor material. On the other hand in N-type region, the epitaxial layer is grown on N-type

substrate. The metal film is used on the P-type layer to provide anode connection to the diode. Similarly, Gold-film layer is coated on N-type to provide cathode connection.

Significance of Gold-film Layer

The gold-film layer on N-type also provides reflection from the bottom surface of the diode. If any significant part of radiated light tends to hit bottom surface then that will be reflected from the bottom surface to the device top surface. This increases LED's efficiency.

Working of LED

The electrons are majority carriers in N-type and holes are majority carriers in P-type. The electrons of N-type are in the conduction band and holes of P-type are in the valence band. The energy level of the Conduction band is higher than the energy level of the Valence band. Thus, if electrons tend to recombine with holes they have to lose some part of the energy to fall in lower energy band.

The electrons can lose their energy either in the form of heat or light. The electrons in Silicon and Germanium lose their energy in the form of heat. Thus, they are not used for LEDs as we want semiconductor in which electrons lose their energy in the form of light.

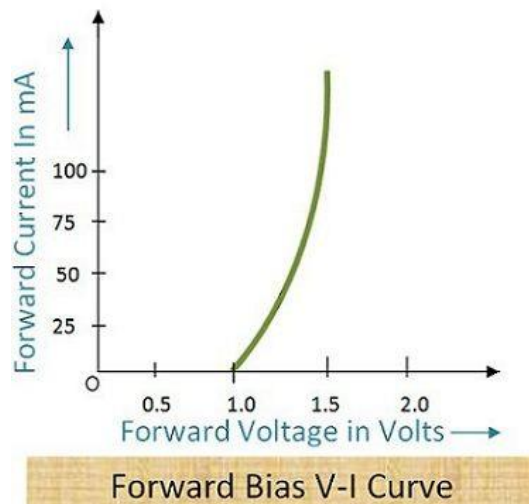
Emission of Photons

Thus, semiconductor compounds such as Gallium Phosphide (GaP), Gallium Arsenide (GaAs), Gallium Arsenide Phosphide (GaAsP) etc. emit light when electrons-holes recombine. The electrons in these compounds lose their energy by emission of photons.

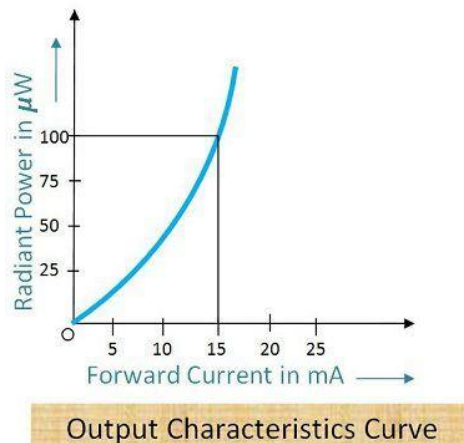
If the semiconductor material is translucent, the light will be emitted from the junction as junction acts as the source of light. LED is operated in forward biased mode only. If it will operate in reverse biased it will get damage as it cannot withstand reverse voltage.

Volt-Ampere Characteristics of LEDs

The characteristics curve of the LED shows that the forward bias of 1 V is sufficient to increase the current exponentially.



The output characteristics curve shows that radiant power of LED is directly proportional the forward current in LED.



Advantages of LED

1. **Temperature Range:** It can be operated over a wide range of temperature ranging from 0°C - 70°C
2. **Switching Time:** The Switching time of LEDs is in order of 1ns. Thus, they are useful in dynamic operations where a large number of arrays are used.
3. **Low Power Consumption:** They consume less power and they can be used even if the dc power supplied is low.
4. **Better Controlling:** The radiant power of LEDs is the function of the current flowing in it. Thus, the light intensity of LED can be controlled easily.
5. **Economical and Reliable:** LEDs are cheap and they possess a high degree of reliability.

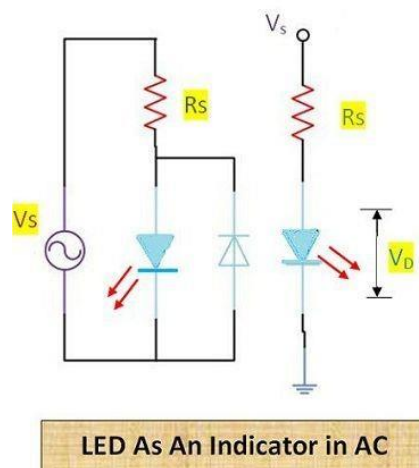
6. Small Size and Portability: They are small in size and can be stacked together for the formation of alphanumeric displays.
7. Higher Efficiency: The efficiency of LEDs to convert power to light energy is 10-50 times greater than that of the tungsten lamp. The response time of LED is 0.1 μ s while in the case of tungsten lamp it is in tens or hundreds of milliseconds.

Disadvantages of LED

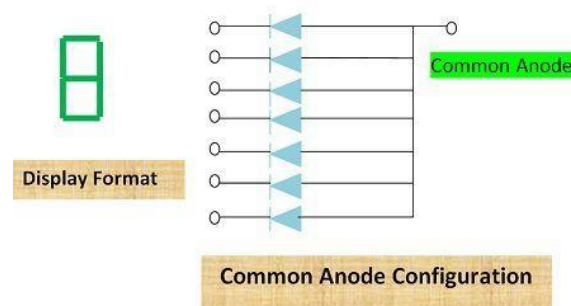
1. Overvoltage or Overcurrent: The LEDs may get damaged when the current is increased beyond a certain limit.
2. Overheating due to radiant power: It gets overheated with an excessive increase in radiant power. This may lead to damage of LED.

Applications of LED

1. Indicator in AC circuit: It can be used as an indicator in AC circuit, but the internal resistance of LED is quite small. Thus, a resistor in series is connected with LED so that the overcurrent can flow through the resistor and can protect LED from getting damaged.



2. Display Panel Indicator: LEDs are used for displaying information processed by electronics circuits. The display format of LED is shown in below diagram.



3. Digital Watches, Calculators & Multimeters: The LEDs which emit visible light are used in digital watches and Calculators for indication purpose.

4. Remote Control Systems & Burglar alarm Systems: Those LEDs which emit invisible infrared light such as GaAs LEDs are used in such applications.

These are the advantages, disadvantages and applications of light emitting diode. LED is a significant optoelectronic device. It is also used in optical fibre communication system.